SciLENS: RL-Driven Autonomous Agents for Scientific Localized Evidence Navigation and Synthesis

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Abstract

Scientific literature synthesis agents increasingly rely on proprietary online services, limiting reproducibility, privacy, and offline deployment. To address this challenge, we introduce **SciLENS** (**Sci**entific **L**ocalized **E**vidence **N**avigation and **S**ynthesis), a fully local autonomous agent framework operating on a dual-tier infrastructure indexing approximately 12 million academic records. SciLENS pioneers the integration of structural visualization as an actionable tool within the reasoning loop, enabling the agent to compress complex citation topologies into validated data-driven charts and thereby mitigate context exhaustion during macro-level synthesis. To train the agent without human annotation, we develop an automated data synthesis pipeline that extracts multi-hop subgraphs from a citation knowledge graph, verified by cross-model consensus among 20 frontier models. The agent is subsequently aligned through a reverse-decomposition rubric strategy that provides fine-grained process rewards for early planning and strict evidence grounding. Evaluations across six scientific benchmarks encompassing standard QA, citation accuracy, factual reasoning, and structural synthesis demonstrate that SciLENS significantly outperforms open-source baselines and achieves performance comparable to GPT-5.2 and Gemini-3.0-pro. Our source code and data are released at https://github.com/LQgdwind/SciLENS.

1 Introduction

As scientific literature grows exponentially, researchers face mounting cognitive challenges in tracing methodological origins and synthesizing macro-level trends (Sun et al., 2024; Zhang et al., 2024). Large language models and autonomous agents are increasingly utilized to automate literature reviews and answer complex scientific inquiries (Singhal et al., 2025; Baek et al., 2025; Zhou et al., 2025; Rouzrokh et al., 2025; Zheng et al., 2025b,a), marking a paradigm shift from static retrieval to agentic workflows (Ma et al., 2024; Vasantharajan, 2025). Three prominent paradigms have emerged to address these challenges: standard retrieval-augmented generation (RAG) pipelines that index localized corpora for granular question answering (Asai et al., 2024; Zheng et al., 2026), web-search agents that dynamically navigate the internet to resolve specific queries (Wu et al., 2025; Li et al., 2025a), and deep-research agents that synthesize multi-document summaries through extended online exploration (Shao et al., 2025; Zhang et al., 2025a). Despite their respective strengths, applying these agents to real-world scientific synthesis reveals several structural limitations that fundamentally constrain their utility.

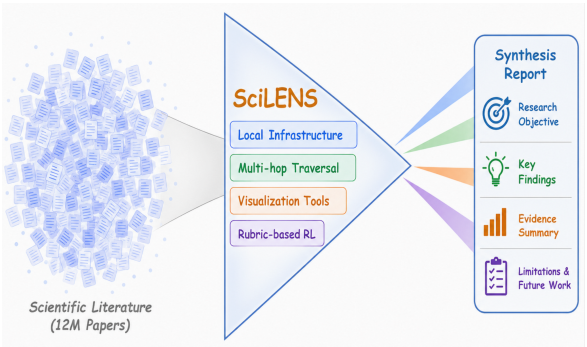


Figure 1: SciLENS operates as a refracting prism: chaotic scientific literature enters the system and is decomposed through four tightly integrated components into structured, multimodal synthesis reports. All processing occurs fully offline with 12M locally indexed papers.

As illustrated in Table 1, existing paradigms exhibit complementary but individually insufficient capabilities when confronted with complex synthesis tasks. Specifically, three critical limitations persist. (1) **Linear text-centric representation bottleneck**: all existing paradigms produce purely textual outputs, leading to context overflow and factual hallucinations when synthesizing macro-

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Table 1: Comparison of existing research paradigms with SciLENS. Each column corresponds to one of the three identified limitations: structural visualization for overcoming text-centric bottlenecks, automated data synthesis with fine-grained rubric-based alignment, and operational independence from fragile web APIs.

Paradigm	Visualization	Auto Synthesis & Alignment	Fully Localized
Standard RAG Pipeline	✗	✗	✓
Web-Search Agents	✗	✗	✗
Deep-Research Agents	✗	✗	✗
SciLENS (Ours)	✓	✓	✓

level trends or comparing complex citation networks (Alansari and Luqman, 2025). No current agent can autonomously compress quantitative topological data into structured visual representations. (2) **Scarcity of training data and absence of fine-grained alignment**: constructing multi-hop interaction trajectories demands prohibitively expensive human annotation, leaving open-source agents inadequately trained (Xu et al., 2024). Moreover, current reinforcement learning strategies rarely supervise implicit logical planning or evidence grounding, and existing benchmarks overlook data-driven visualization and large-scale summarization capabilities (Vasantharajan, 2025). (3) **Dependency on fragile online infrastructure**: web-search and deep-research agents rely on external search APIs, exposing them to high latency, restrictive rate limits, and network failures that hinder rapid, deterministic access to large-scale corpora (Shao et al., 2025; Yao et al., 2026).

To address these limitations, we propose **SciLENS** (**Sci**entific **L**ocalized **E**vidence **N**avigation and **S**ynthesis), a fully localized autonomous agent framework for complex scientific reasoning and structural synthesis. Our central hypothesis is that a fully localized agent equipped with multi-hop citation traversal, structural visualization tools, and rubric-based reinforcement learning can achieve scientific synthesis performance comparable to frontier proprietary models without online dependency.

To realize this hypothesis, SciLENS is designed around three key components. First, to overcome text-centric representation bottlenecks, we integrate structural visualization tools into the agent reasoning loop, enabling the agent to generate validated chart schemas that compress complex citation topologies into high-density visual summaries (Section 2.2). Second, to address training data scarcity and fine-grained alignment, we develop an automated data synthesis pipeline that extracts multi-hop subgraphs from a citation knowledge graph and generates question-answer pairs verified through cross-model consensus (Section 3.2). We further align the agent with rubric-based reinforcement learning, using reverse-decomposition rubrics to supervise early planning and strict evidence grounding (Section 3.3). Finally, to ensure fully local and robust deployment, SciLENS operates on a dual-tier local infrastructure that combines MongoDB for metadata management with distributed FAISS for dense retrieval, indexing approximately 12 million academic records for subsecond access (Section 2.1).

In summary, our contributions are fourfold:

(1) **Visualization-Augmented Scientific Reasoning.** We integrate structural visualization as an actionable tool within the autonomous reasoning loop, enabling the agent to summarize complex citation topologies and quantitative trends as validated data-driven charts. This design alleviates the limitations of text-only reasoning by producing structured visual outputs that reduce context burden during macro-level synthesis.

(2) **Citation-Graph Data Synthesis and Rubric-Based Alignment.**We develop an automated data synthesis pipeline that extracts citation subgraphs via random walks and generates multi-hop reasoning trajectories verified through cross-model consensus, eliminating the need for human annotation. We then align the agent with reverse-decomposition rubrics that reward early decomposition and strict evidence grounding.

(3) **Fully Localized Agentic Infrastructure.** We further build a fully local infrastructure that indexes approximately 12 million academic records and supports semantic retrieval, citation-graph traversal, and structural visualization through an offline research toolbox.

(4) **Comprehensive Evaluation.** We conduct extensive evaluations across six scientific benchmarks spanning reading comprehension, citation accuracy, factual reasoning, and structural synthesis, demonstrating that SciLENS outperforms open-source baselines and approaches frontier proprietary models.

2 Agent Architecture and Visualization-Augmented Reasoning

To perform complex academic reasoning and macro-level synthesis without the latency, instability, and API constraints of external web search engines, we design **SciLENS**, a fully localized and autonomous agent framework. The framework

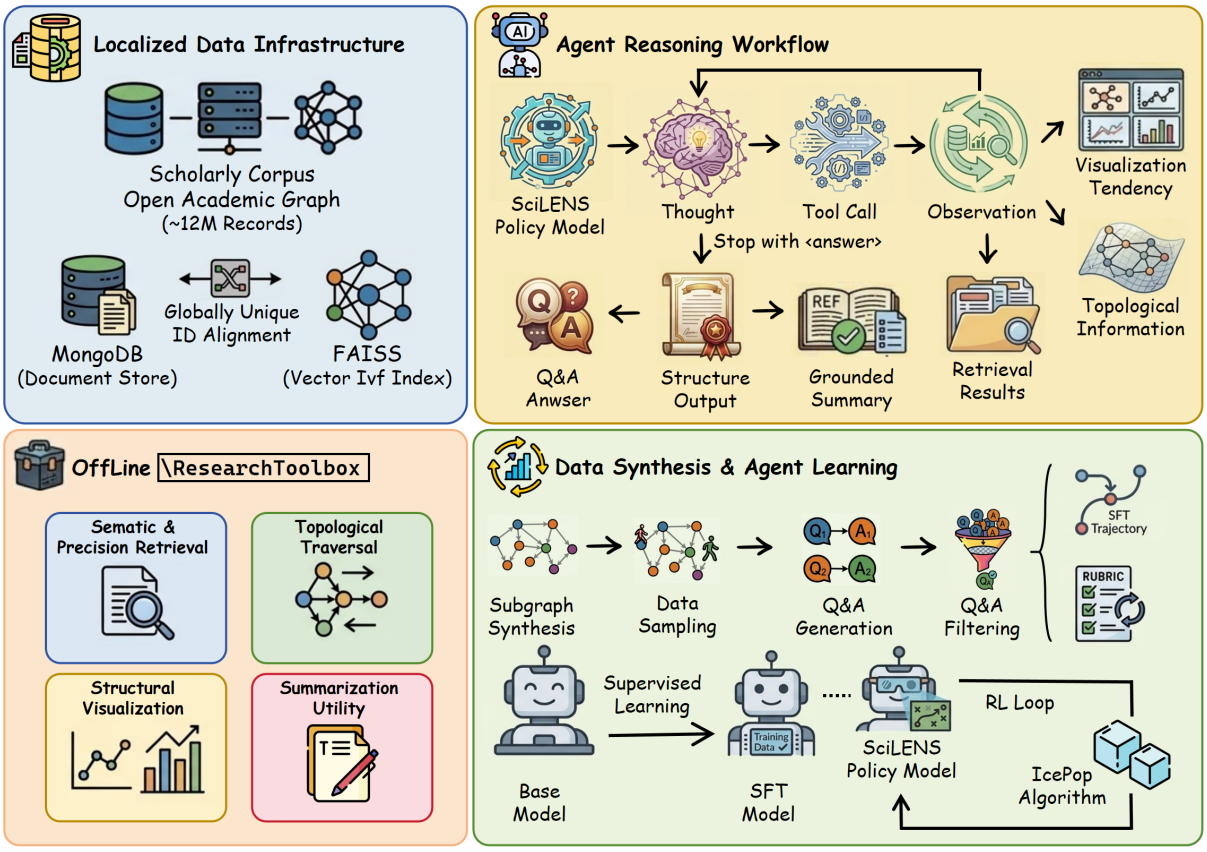


Figure 2: The SciLENS architecture integrates four core components. A localized dual-tier infrastructure ensures deterministic access to a massive scholarly corpus. Utilizing this foundation, an offline research toolbox empowers the agent with semantic retrieval, topological traversal, and structural visualization capabilities orchestrated via a continuous reasoning loop. Finally, an automated data synthesis pipeline generates complex trajectories to explicitly align the policy model through supervised fine-tuning and rubric-based reinforcement learning.

combines a high-throughput dual-tier infrastructure with a visualization-augmented offline tool suite, enabling deterministic data access and multimodal structured outputs beyond linear text generation.

2.1 Localized Dual-Tier Infrastructure

The foundation of SciLENS is a persistent, fully localized storage backend that indexes approximately 12 million cleaned academic records derived from the Open Academic Graph corpus (Zhang et al., 2022). To achieve sub-second retrieval latency while maintaining rich metadata access, we construct a dual-tier architecture coupling a MongoDB document store with a distributed FAISS vector index. The MongoDB tier handles exact-match queries, keyword-based BM25 (Robertson and Zaragoza, 2009) searches, and citation relationship tracking, while the FAISS tier, built across independent GPU shards using Qwen3-Embedding-8B (Zhang et al., 2025b), enables distributed dense retrieval. The two tiers are coupled through a shard-based identifier alignment scheme, where each paper is assigned a globally unique integer computed from its shard index and within-shard offset, allowing nearest-neighbor vector results to be resolved to full paper metadata via a single indexed database lookup. Instead of relying on slow, synchronous online web searches, our agent interacts exclusively with this local infrastructure, thereby eliminating network-induced latency and failure modes. The detailed system configurations, including chunk-based parallel ingestion strategies, identifier alignment, and API batching parameters, are provided in Appendix A.

2.2 Visualization-Augmented Offline Tool Suite

Existing research agents operate exclusively through linear text generation, which inevitably leads to context exhaustion and factual hallucinations when synthesizing macro-level academic trends across dozens of papers (Alansari and Luqman, 2025). To address this bottleneck, we introduce a key algorithmic insight: agents autonomously invoke visualization tools during reasoning to compress high-dimensional topology into chart schemas, bypassing inefficient textual encoding. This transforms the agent from a purely text-

centric reasoner into a multimodal synthesizer capable of producing integrated text-and-chart academic reports.

To operationalize this insight, we design a unified RESEARCHTOOLBOX with twelve tools across four functional categories. Semantic retrieval tools support dense embedding search, keyword lookup, and metadata extraction, while topological traversal tools enable multi-hop graph exploration and shortest-path tracing over citation networks. Our novel visualization tools enable the agent to emit validated schemas for standard charts (e.g., line, bar, scatter), strictly enforcing dimensional and numeric consistency prior to rendering. A dedicated summarization utility assists in distilling extensive academic abstracts to optimize context window utilization. The agent accesses all capabilities via structured JSON payloads and receives observations directly into its reasoning chain. The formal tool specifications, precise input schemas, and operational boundaries are detailed in Appendix B.

2.3 Agent Reasoning Workflow

Equipped with the RESEARCHTOOLBOX, the SciLENS agent operates in a continuous thought, action, and observation loop governed by a rigorous system prompt. This foundational instruction set is universally applied across both the training and inference phases to ensure strict adherence to tool invocation schemas and behavioral constraints. The complete directives establishing these operational boundaries are detailed in Appendix C. Complex queries are processed by the agent through sequential seed document retrieval, topology-based multi-hop evidence traversal, and autonomous metric structuring for visual trend analysis. To ensure robust execution, we adopt a dual-tier error recovery mechanism: formatting errors trigger contextualized feedback for autonomous correction, while missing tool calls or answer tags lead to immediate re-rolls. The agent then integrates textual and structural evidence to generate coherent academic synthesis, reducing factual hallucinations.

3 Citation-Graph Data Synthesis and Rubric-Based Alignment

Training autonomous agents for complex scientific reasoning requires high-quality, multi-hop interaction trajectories. However, such data is notoriously scarce and prohibitively expensive to annotate manually. To address this bottleneck, we propose an automated, logic-driven data synthesis pipeline that extracts cohesive subgraphs from academic literature to formulate complex question-answer pairs. We then utilize the evolutionary history of these questions to align the agent through a novel rubric-based reinforcement learning strategy. The overall training framework is illustrated in Figure 2.

3.1 Topological Subgraph Sampling

To generate locally coherent contexts for downstream reasoning synthesis, we first extract highly interconnected document clusters from the Open Academic Graph corpus (Zhang et al., 2022). Rather than relying on static semantic similarity, we capture the actual citation topology by executing random walk explorations. Starting from a uniformly sampled seed publication, a walker proceeds for thirty steps, transitioning to randomly selected forward or backward citation neighbors. We execute ten independent walks per subgraph to form a connected local network. A total of thirty thousand subgraphs are generated in this manner. Each subgraph serves as a cohesive factual grounding cluster for subsequent question generation, ensuring that the synthesized tasks reflect real-world academic literature structures. The detailed node-filtering heuristics and streaming deduplication processes applied prior to sampling are provided in Appendix D.

3.2 Logic-Driven QA Synthesis and Verification

Starting from topological subgraphs, we synthesize multi-hop question-answer pairs by compounding factoid base questions with sampled relational links. Difficulty is ensured via a self-consistency mechanism where the model must fail to resolve the composite question using only atomic facts. Subsequently, each candidate instance undergoes strict cross-model verification utilizing a diverse pool of twenty frontier language models. For each candidate, we implement a stochastic verification protocol where 4 verifiers are drawn uniformly at random from the pool to independently render binary judgments on factual correctness and answer uniqueness via autonomous retrieval queries. A candidate QA pair is retained if and only if all sampled verifiers reach an absolute consensus:

$$\text{keep}(q,a) = \prod_{m \in \mathcal{S}} \mathbf{1}[\text{corr}_m(q,a) \wedge \text{uniq}_m(q,a)]\,, \tag{1}$$

where $\mathcal{S} \subset \{1, \ldots, 20\}$ such that $|\mathcal{S}| = 4$ represents the random subset of verifiers, while corr_m and uniq_m denote the correctness and uniqueness judgments of verifier m. This probabilistic filtering ensures high precision training data while maintaining broad model diversity. The exhaustive list of constituent models and the detailed synthesis procedures are provided in Appendix D.

3.3 Multi-Step Multi-Tool Agent Learning

With the high-quality QA instances synthesized, we develop a two stage training paradigm to empower SciLENS with tool use proficiency and logical rigor.

Supervised Fine-Tuning via Distillation. Recent studies examine how information transfer can be controlled in LLM distillation, from distillation resistance to filtered and reweighted on-policy objectives (Fang et al., 2026; Li et al., 2026). In the first stage, we perform knowledge distillation from DeepSeek-V3.2 (Liu et al., 2025a). The teacher model interacts with the RESEARCHTOOLBOX to generate approximately twenty thousand complete interaction trajectories. To avoid interference from external feedback during learning and enhance the capability of the model to couple reasoning and action steps, we mask out loss contributions from tool observations. Given a task context c and a trajectory $H = (x_1, x_2, \ldots, x_{|H|})$ where each x_i represents either a reasoning step, a tool action, or an observation o, the supervised loss function is computed as follows:

$$\mathcal{L}_{\text{SFT}} = -\frac{\sum_{i=1}^{|H|} \mathbb{I}[x_i \neq o] \cdot \log \pi_\theta(x_i | c, x_{<i})}{\sum_{i=1}^{|H|} \mathbb{I}[x_i \neq o]}. \tag{2}$$

This masking mechanism preserves original reasoning capabilities while teaching a robust behavioral paradigm of alternating deliberate reasoning with tool execution.

Rubric-Based Reinforcement Learning. During the reinforcement learning phase, we optimize the policy model utilizing the IcePop algorithm (Team et al., 2025a). IcePop uses a decoupled behavior policy and selective token masking to stabilize exploration and avoid policy collapse. The full objective is provided in Appendix F. Training trajectories are collected via agentic ReAct rollouts. The rigorous formatting constraints and structural tags governing these rollouts are universally defined by the system prompt, with complete execution details deferred to Appendix C. The reward design closely aligns optimization across rollouts. Instead of a single scalar reward, we adopt a multi-dimensional framework combining Reinforcement Learning from Verifiable Rewards for final answer correctness with semantic rewards for planning, reverse decomposition, and evidence grounding. Prompting a judge model generates an instance-specific rubric comprising N weighted criteria for each training query:

$$\mathcal{R} = \{(d_i, w_i)\}_{i=1}^{N}, \quad \sum_{i=1}^{N} w_i = 100, \tag{3}$$

where d_i is a natural language description of the verification criterion and w_i is its normalized importance weight. Finally, we compute a composite scalar reward to evaluate the entire trajectory. The reward integrates a binary format score S_{format} ensuring strict conformity to required schemas, alongside a semantic answer score S_{rubric} evaluated against the criteria in $\mathcal{R}$. The composite reward function is defined as:

$$R = \lambda \cdot S_{\text{format}} + (1 - \lambda) \cdot S_{\text{rubric}}. \tag{4}$$

This fine-grained alignment guarantees that SciLENS prioritizes structured early planning and meticulous evidence grounding over superficial text generation. Comprehensive construction methodologies for these rubrics are provided in Appendix G.

4 Benchmarks and Multidimensional Evaluation

To rigorously assess the capabilities of SciLENS, we establish a comprehensive evaluation framework encompassing six benchmarks across two complementary categories. The established evaluations utilize SciFact (Wadden et al., 2020), PubMedQA (Jin et al., 2019), QASA (Lee et al., 2023), and ScholarQA-CS (Singh et al., 2025) to measure correctness and citation accuracy. Beyond standard reading comprehension, we evaluate on Structural Synthesis (SSB) and Scientific Fact and Reasoning (SciFR), which assess visual consistency, query alignment, semantic fidelity, and evidence attribution. Both benchmarks are constructed via automated multi-hop subgraph extraction from computer science, physics, and biomedicine, yielding 500 verified instances each, with strict train–test disjointness enforced at the subgraph, seed-paper,

and instance levels (Appendix E). For reproducibility, all evaluations use a locally deployed Qwen3-30B-A3B judge with zero temperature. Full evaluation protocols, metric definitions, and instance-level rubrics are provided in Appendix H.

5 Experiment

5.1 Experimental Setup

To establish competitive performance upper bounds and evaluate the capability gap across diverse architectures, we compare SciLENS against a comprehensive suite of state-of-the-art proprietary models and leading open-source frameworks, spanning general-purpose reasoning backbones, specialized deep research agents, and exploratory web agents. The exhaustive list of evaluated baselines and specific training configurations are documented in Appendix I.

A critical design principle of our evaluation is the **unified toolbox protocol**: all evaluated models, including proprietary systems such as GPT-5.2 and Claude-4.5-Sonnet, are equipped with the identical offline RESEARCHTOOLBOX and evaluated against the same 12-million-paper local database. No model uses web search during evaluation. This design ensures that all performance differences reported in Tables 2a and 2b are attributable solely to model capabilities and training methodology, not to infrastructure or tool advantages. In particular, the Qwen3-30B-A3B (Base + Tools) row in our tables represents the zero-shot base model given full access to the entire RESEARCHTOOLBOX without any SciLENS training, serving as a controlled reference that isolates the contribution of our training pipeline from the underlying tool access.

5.2 Main Results

We conduct extensive evaluations against state-of-the-art proprietary and leading open-source models, with results reported in Tables 2a and 2b. The SciLENS rows in the tables detail the full training pipeline, including the base model with full RESEARCHTOOLBOX access, the SFT variant, and the final RL model. On four established benchmarks, SciLENS-RL achieves the highest correctness scores on QASA (47.62), SciFact (88.94), and PubMedQA (77.53), and attains the strongest citation accuracy across all datasets, demonstrating robust factual grounding on external evaluations. On SSB and SciFR, the Qwen3-30B-A3B (Base + Tools) row shows that tool access alone yields only 0.13 and 0.17 respectively, establishing that these tasks cannot be solved without alignment. Supervised fine-tuning dramatically closes this gap, and the reinforcement learning variant achieves the strongest overall scores, outperforming all open-source baselines and achieving performance comparable to GPT-5.2 and Gemini-3.0-pro. An extended analysis of performance behaviors across these benchmarks is provided in Appendix J.1.

5.3 Ablation Studies

We conduct two complementary ablation studies under a unified toolbox protocol to isolate the contribution of each component. Since they measure orthogonal dimensions, their magnitudes are not directly comparable. The tool-necessity ablation (Appendix J) shows that complex scientific synthesis cannot be solved without retrieval: removing retrieval and graph tools reduces the agent to a closed-book setting, leading to consistent performance collapse across benchmarks. This confirms that our benchmarks require multi-hop evidence gathering rather than parametric recall. In contrast, the RL ablation measures algorithmic gains over a strong SFT baseline with identical tool access across settings; thus, observed improvements reflect purely training-driven contributions.

As shown in Tables 2a and 2b, Qwen3-30B-A3B (Base + Tools) demonstrates that tool access alone is insufficient for scientific synthesis, yielding only 0.1324 on SSB and 0.1654 on SciFR. This outcome is expected, as the Qwen3-30B-A3B base model inherently lacks robust multi-turn tool orchestration capabilities prior to SFT alignment, frequently failing to parse tool responses or chain sequential retrievals coherently. Supervised fine-tuning substantially improves performance across all benchmarks, while rubric-based reinforcement learning further yields consistent gains, including notable improvements on ScholarQA-CS (+7.46) and SciFR (+0.098). Table 3 provides a detailed breakdown of RL components. Answer RLVR provides the core correctness signal; its removal leads to substantial performance drops across all benchmarks, indicating that outcome-based supervision is necessary for effective policy optimization. The planning and grounding rubrics each yield consistent gains, particularly at the upper performance range. Notably, removing planning rubrics substantially degrades SSB performance ($0.7607 \rightarrow 0.6855$), indicating that process-level supervision is essential for complex structural synthe-

Table 2: Performance comparison across all benchmarks. All models are evaluated with the identical local RESEARCHTOOLBOX and 12M-paper database; no model uses web search (Section 5.1). All reported scores represent the mean ± standard deviation across five independent evaluation runs ($p < 0.05$). **Bold** indicates the best performance of SciLENS; *italic* indicates the best baseline performance in each section.

(a) Structural Synthesis Benchmark (SSB) and Scientific Fact and Reasoning (SciFR)

Model	Structural Synthesis Benchmark				Scientific Fact and Reasoning			
	ACS	QAS	FGS	Overall	ASF	QIS	EGQ	Overall
Closed-source LLMs								
Claude-4-Sonnet	$0.3007_{\pm.0308}$	$0.3003_{\pm.0167}$	$0.3020_{\pm.0139}$	$0.3009_{\pm.0455}$	$0.2922_{\pm.0380}$	$0.3112_{\pm.0496}$	$0.5686_{\pm.0378}$	$0.3640_{\pm.0267}$
Claude-4.5-Sonnet	$0.4552_{\pm.0270}$	$0.5651_{\pm.0467}$	$0.3845_{\pm.0481}$	$0.4959_{\pm.0388}$	$0.2328_{\pm.0128}$	$0.3399_{\pm.0476}$	$0.4190_{\pm.0206}$	$0.3060_{\pm.0443}$
Gemini-2.5-pro	$0.2641_{\pm.0169}$	$0.3794_{\pm.0518}$	$0.3933_{\pm.0279}$	$0.3246_{\pm.0527}$	$0.4373_{\pm.0238}$	$0.7370_{\pm.0444}$	$0.5978_{\pm.0393}$	$0.5562_{\pm.0535}$
Gemini-3.0-pro	$\mathit{0.5991}_{\pm.0580}$	$\mathit{0.7734}_{\pm.0493}$	$\mathit{0.6868}_{\pm.0306}$	$\mathit{0.6707}_{\pm.0419}$	$0.4954_{\pm.0366}$	$0.7340_{\pm.0396}$	$0.5956_{\pm.0255}$	$0.5839_{\pm.0354}$
GPT-5	$0.5857_{\pm.0428}$	$0.7340_{\pm.0456}$	$0.6443_{\pm.0391}$	$0.6437_{\pm.0499}$	$0.5640_{\pm.0465}$	$0.6585_{\pm.0271}$	$0.5601_{\pm.0357}$	$0.5885_{\pm.0439}$
GPT-5.1	$0.5581_{\pm.0436}$	$0.7069_{\pm.0544}$	$0.6313_{\pm.0489}$	$0.6193_{\pm.0523}$	$0.6133_{\pm.0539}$	$0.7189_{\pm.0421}$	$\mathit{0.6035}_{\pm.0459}$	$0.6394_{\pm.0266}$
GPT-5.2	$0.5680_{\pm.0452}$	$0.7124_{\pm.0529}$	$0.6321_{\pm.0304}$	$0.6263_{\pm.0460}$	$\mathit{0.6349}_{\pm.0319}$	$\mathit{0.7717}_{\pm.0313}$	$0.5772_{\pm.0479}$	$\mathit{0.6578}_{\pm.0578}$
OpenAI-DeepResearch	$0.5966_{\pm.0553}$	$0.7439_{\pm.0299}$	$0.6617_{\pm.0392}$	$0.6564_{\pm.0402}$	$0.5800_{\pm.0469}$	$0.7223_{\pm.0478}$	$0.5696_{\pm.0528}$	$0.6158_{\pm.0339}$
Perplexity-DeepResearch	$0.4743_{\pm.0418}$	$0.5719_{\pm.0285}$	$0.5644_{\pm.0348}$	$0.5232_{\pm.0492}$	$0.5676_{\pm.0356}$	$0.6844_{\pm.0541}$	$0.5582_{\pm.0430}$	$0.5969_{\pm.0508}$
Open-source LLMs								
A-Searcher	$0.4343_{\pm.0571}$	$0.6716_{\pm.0394}$	$0.5369_{\pm.0386}$	$0.5291_{\pm.0221}$	$0.4942_{\pm.0375}$	$0.7501_{\pm.0358}$	$0.5097_{\pm.0410}$	$0.5672_{\pm.0547}$
DR-Tulu	$0.3450_{\pm.0153}$	$0.5778_{\pm.0400}$	$0.4743_{\pm.0320}$	$0.4433_{\pm.0194}$	$0.4046_{\pm.0483}$	$0.7906_{\pm.0453}$	$0.4100_{\pm.0323}$	$0.5101_{\pm.0205}$
Mirothinker-v1.5-30B	$0.3930_{\pm.0316}$	$0.5672_{\pm.0283}$	$0.4922_{\pm.0203}$	$0.4671_{\pm.0377}$	$0.4896_{\pm.0448}$	$0.7745_{\pm.0519}$	$\mathit{0.5316}_{\pm.0230}$	$0.5764_{\pm.0249}$
Tongyi-Deepresearch	$0.3541_{\pm.0234}$	$0.5080_{\pm.0291}$	$0.4006_{\pm.0429}$	$0.4123_{\pm.0262}$	$0.2749_{\pm.0497}$	$0.4564_{\pm.0440}$	$0.4372_{\pm.0233}$	$0.3624_{\pm.0362}$
Web-Thinker	$\mathit{0.4936}_{\pm.0310}$	$\mathit{0.8156}_{\pm.0475}$	$0.5314_{\pm.0562}$	$\mathit{0.6028}_{\pm.0470}$	$0.2044_{\pm.0141}$	$0.3786_{\pm.0403}$	$0.3234_{\pm.0324}$	$0.2798_{\pm.0208}$
Web-Dancer	$0.4902_{\pm.0236}$	$0.7726_{\pm.0551}$	$\mathit{0.5788}_{\pm.0574}$	$0.5966_{\pm.0300}$	$\mathit{0.5315}_{\pm.0342}$	$0.8094_{\pm.0390}$	$0.5232_{\pm.0281}$	$\mathit{0.6042}_{\pm.0557}$
Web-Explorer	$0.2792_{\pm.0201}$	$0.3742_{\pm.0415}$	$0.3628_{\pm.0335}$	$0.3250_{\pm.0250}$	$0.4017_{\pm.0404}$	$0.5582_{\pm.0572}$	$0.5173_{\pm.0364}$	$0.4722_{\pm.0360}$
Web-Shaper	$0.2445_{\pm.0487}$	$0.3652_{\pm.0442}$	$0.2754_{\pm.0450}$	$0.2888_{\pm.0413}$	$0.2224_{\pm.0217}$	$0.3574_{\pm.0173}$	$0.4290_{\pm.0287}$	$0.3087_{\pm.0239}$
OpenScholar-8B	$0.1845_{\pm.0121}$	$0.2102_{\pm.0423}$	$0.1956_{\pm.0156}$	$0.1967_{\pm.0331}$	$0.3850_{\pm.0383}$	$0.4621_{\pm.0361}$	$0.3540_{\pm.0231}$	$0.4003_{\pm.0471}$
O-Researcher-72B-rl	$0.4742_{\pm.0485}$	$0.7599_{\pm.0355}$	$0.5501_{\pm.0525}$	$0.5796_{\pm.0384}$	$0.4726_{\pm.0274}$	$\mathit{0.8676}_{\pm.0466}$	$0.5055_{\pm.0225}$	$0.5860_{\pm.0363}$
O-Researcher-72B-sft	$0.4705_{\pm.0273}$	$0.7557_{\pm.0387}$	$0.5412_{\pm.0243}$	$0.5749_{\pm.0449}$	$0.3562_{\pm.0252}$	$0.6834_{\pm.0426}$	$0.4969_{\pm.0224}$	$0.4784_{\pm.0517}$
SciLENS (Ours)								
Qwen3-30B-A3B (Base + Tools)	$0.1232_{\pm.0309}$	$0.1419_{\pm.0130}$	$0.1401_{\pm.0432}$	$0.1324_{\pm.0277}$	$0.1523_{\pm.0154}$	$0.1834_{\pm.0237}$	$0.1605_{\pm.0119}$	$0.1654_{\pm.0317}$
SciLENS-30B (SFT)	$0.6336_{\pm.0269}$	$0.9054_{\pm.0382}$	$0.6820_{\pm.0501}$	$0.7290_{\pm.0398}$	$0.6301_{\pm.0472}$	$0.7863_{\pm.0537}$	$0.5834_{\pm.0567}$	$0.6611_{\pm.0434}$
SciLENS-30B (RL)	$\mathbf{0.6811}_{\pm.0579}$	$\mathbf{0.9127}_{\pm.0534}$	$\mathbf{0.7138}_{\pm.0370}$	$\mathbf{0.7607}_{\pm.0569}$	$\mathbf{0.7395}_{\pm.0326}$	$\mathbf{0.8364}_{\pm.0512}$	$\mathbf{0.7144}_{\pm.0416}$	$\mathbf{0.7594}_{\pm.0575}$

(b) Standard Scientific Reasoning Benchmarks

Model	QASA		SciFact		PubMedQA		ScholarQA CS	
	Corr	Cite	Corr	Cite	Corr	Cite	Corr	Cite
Closed-source LLMs								
Claude-4-Sonnet	$33.42_{\pm0.66}$	$31.41_{\pm1.94}$	$60.58_{\pm2.52}$	$42.39_{\pm2.24}$	$50.96_{\pm2.45}$	$45.32_{\pm2.39}$	$39.52_{\pm1.67}$	$23.18_{\pm1.50}$
Claude-4.5-Sonnet	$\mathit{46.27}_{\pm2.54}$	$29.03_{\pm2.22}$	$\mathit{86.06}_{\pm1.33}$	$40.33_{\pm1.55}$	$\mathit{73.90}_{\pm0.87}$	$43.29_{\pm2.08}$	$67.92_{\pm1.07}$	$25.64_{\pm1.22}$
Gemini-2.5-pro	$30.49_{\pm0.98}$	$27.66_{\pm1.00}$	$47.60_{\pm1.35}$	$49.59_{\pm1.66}$	$46.62_{\pm2.62}$	$69.84_{\pm1.93}$	$46.44_{\pm2.34}$	$\mathit{60.83}_{\pm1.25}$
Gemini-3.0-pro	$31.10_{\pm1.82}$	$33.20_{\pm1.72}$	$69.71_{\pm2.88}$	$\mathit{67.13}_{\pm2.21}$	$48.87_{\pm0.84}$	$\mathit{72.13}_{\pm1.16}$	$45.73_{\pm1.68}$	$49.46_{\pm2.82}$
GPT-5	$44.14_{\pm0.77}$	$16.83_{\pm1.11}$	$82.21_{\pm2.16}$	$50.03_{\pm2.89}$	$69.75_{\pm2.57}$	$58.71_{\pm2.18}$	$68.34_{\pm1.43}$	$20.21_{\pm1.08}$
GPT-5.1	$41.08_{\pm2.19}$	$\mathit{35.65}_{\pm2.01}$	$83.65_{\pm1.54}$	$48.53_{\pm1.95}$	$68.62_{\pm1.17}$	$35.14_{\pm1.14}$	$\mathit{73.27}_{\pm1.26}$	$28.75_{\pm1.51}$
GPT-5.2	$44.22_{\pm1.85}$	$26.79_{\pm1.74}$	$80.29_{\pm2.04}$	$33.98_{\pm1.63}$	$65.95_{\pm1.52}$	$48.19_{\pm0.94}$	$71.88_{\pm2.12}$	$32.91_{\pm1.19}$
OpenAI-DeepResearch	$38.92_{\pm1.12}$	$17.02_{\pm1.75}$	$73.56_{\pm1.28}$	$42.94_{\pm2.37}$	$60.02_{\pm1.41}$	$55.20_{\pm1.27}$	$66.20_{\pm0.88}$	$24.32_{\pm0.92}$
Perplexity-DeepResearch	$28.57_{\pm2.25}$	$14.44_{\pm1.70}$	$55.77_{\pm1.61}$	$25.05_{\pm1.99}$	$49.77_{\pm1.01}$	$36.95_{\pm2.67}$	$65.41_{\pm2.30}$	$28.10_{\pm0.67}$
Open-source LLMs								
A-Searcher	$31.97_{\pm1.87}$	$32.52_{\pm1.39}$	$59.13_{\pm0.83}$	$54.02_{\pm1.96}$	$44.48_{\pm2.78}$	$65.65_{\pm2.61}$	$40.61_{\pm2.47}$	$47.87_{\pm1.29}$
DR-Tulu	$25.26_{\pm0.68}$	$8.10_{\pm1.64}$	$70.48_{\pm2.56}$	$26.02_{\pm2.46}$	$59.79_{\pm2.31}$	$30.50_{\pm2.23}$	$51.10_{\pm2.85}$	$\mathit{72.34}_{\pm1.34}$
Mirothinker-v1.5-30B	$\mathit{42.95}_{\pm1.76}$	$33.59_{\pm1.09}$	$\mathit{80.75}_{\pm1.88}$	$58.52_{\pm2.35}$	$\mathit{69.80}_{\pm2.84}$	$62.92_{\pm2.20}$	$\mathit{66.21}_{\pm1.65}$	$39.45_{\pm1.13}$
Tongyi-Deepresearch	$34.18_{\pm1.45}$	$34.04_{\pm1.42}$	$61.25_{\pm2.80}$	$\mathit{70.30}_{\pm1.81}$	$51.44_{\pm2.43}$	$62.01_{\pm1.58}$	$40.17_{\pm2.42}$	$46.73_{\pm0.72}$
Web-Thinker	$10.66_{\pm0.55}$	$20.42_{\pm1.79}$	$33.65_{\pm1.97}$	$49.46_{\pm0.93}$	$10.20_{\pm1.69}$	$62.16_{\pm1.46}$	$14.55_{\pm1.59}$	$26.52_{\pm2.29}$
Web-Dancer	$35.49_{\pm1.77}$	$27.94_{\pm1.92}$	$67.79_{\pm2.68}$	$51.53_{\pm1.62}$	$58.78_{\pm1.47}$	$59.11_{\pm2.71}$	$51.94_{\pm2.49}$	$33.47_{\pm2.14}$
Web-Explorer	$27.68_{\pm2.06}$	$15.59_{\pm1.05}$	$57.08_{\pm1.86}$	$26.01_{\pm2.28}$	$51.72_{\pm1.31}$	$21.03_{\pm1.20}$	$36.90_{\pm2.76}$	$26.03_{\pm0.76}$
Web-Shaper	$18.75_{\pm0.79}$	$\mathit{49.69}_{\pm1.10}$	$24.04_{\pm1.06}$	$68.80_{\pm2.95}$	$19.47_{\pm2.15}$	$\mathit{74.05}_{\pm2.02}$	$22.43_{\pm1.83}$	$50.84_{\pm2.65}$
OpenScholar-8B	$24.56_{\pm1.78}$	$38.79_{\pm1.73}$	$77.52_{\pm2.09}$	$51.28_{\pm0.89}$	$65.49_{\pm2.48}$	$60.32_{\pm2.73}$	$50.28_{\pm0.97}$	$54.36_{\pm2.36}$
O-Researcher-72B-sft	$28.99_{\pm1.90}$	$18.40_{\pm1.53}$	$48.56_{\pm0.85}$	$40.83_{\pm2.07}$	$27.09_{\pm0.82}$	$29.84_{\pm2.75}$	$44.29_{\pm2.66}$	$27.62_{\pm2.55}$
O-Researcher-72B-rl	$35.51_{\pm2.05}$	$28.78_{\pm0.65}$	$74.23_{\pm2.38}$	$25.04_{\pm0.95}$	$55.52_{\pm2.60}$	$33.51_{\pm1.15}$	$59.34_{\pm2.63}$	$71.22_{\pm2.03}$
SciLENS (Ours)								
Qwen3-30B-A3B (Base + Tools)	$5.90_{\pm0.53}$	$2.44_{\pm0.57}$	$14.42_{\pm1.48}$	$10.21_{\pm0.54}$	$14.95_{\pm1.03}$	$7.71_{\pm1.37}$	$5.44_{\pm2.58}$	$5.68_{\pm0.69}$
SciLENS-30B (SFT)	$46.43_{\pm2.87}$	$50.20_{\pm1.32}$	$85.87_{\pm2.50}$	$79.03_{\pm1.80}$	$73.84_{\pm2.40}$	$72.97_{\pm2.10}$	$66.52_{\pm2.90}$	$75.03_{\pm2.70}$
SciLENS-30B (RL)	$\mathbf{47.62}_{\pm2.27}$	$\mathbf{52.16}_{\pm2.91}$	$\mathbf{88.94}_{\pm1.40}$	$\mathbf{83.72}_{\pm2.41}$	$\mathbf{77.53}_{\pm2.72}$	$\mathbf{76.54}_{\pm2.33}$	$\mathbf{73.98}_{\pm1.24}$	$\mathbf{76.82}_{\pm0.90}$

Table 3: RL component ablation of SciLENS. All configurations use the full RESEARCHTOOLBOX and SFT initialization. Metrics represent Correctness for standard benchmarks and Overall for SciFR and SSB ($p < 0.05$).

Configuration	QASA	SciFact	PubMedQA	ScholarQA	SciFR	SSB
SciLENS (Full RL)	**47.62**	**88.94**	**77.53**	**73.98**	**0.7594**	**0.7607**
w/o Answer RLVR	41.20	72.45	64.33	62.10	0.7115	0.7350
w/o Planning Rubrics	47.10	86.80	74.90	70.40	0.7310	0.6855
w/o Grounding Rubrics	46.50	86.17	75.80	72.40	0.7321	0.7020

sis.A complementary tool necessity ablation (Appendix J) confirms that retrieval and visualization tools are essential prerequisites: removing visualization tools alone causes SSB to drop sharply from 0.7607 to 0.5215 while standard QA benchmarks remain comparatively stable, directly validating the visualization-as-tool contribution for structural synthesis.

5.4 Extended Analytical Experiments

We further analyze SciLENS through long-context management (Appendix J.2), tool utilization dynamics (Appendix J.3), and RL convergence analysis (Appendix J.4). Qualitative case studies and error analyses are provided in Appendix K.

6 Related Work

6.1 Autonomous Online Search Agents

LLMs have advanced biomedical (Luo et al., 2022; Chen et al., 2023; Yang et al., 2022; Tu et al., 2024), medical (Zhou et al., 2023; Wei et al., 2024; Singhal et al., 2025), geoscience (Deng et al., 2024), astronomy (Nguyen et al., 2023), and multidisciplinary domains (Sun et al., 2024; Zhang et al., 2024; Cai et al., 2024). Beyond static queries, models now automate code generation (Gero et al., 2022; Du et al., 2024) and research ideation (Baek et al., 2025; Kumar et al., 2025), culminating in autonomous agents (Wu et al., 2025; Liu et al., 2025b) that execute complex literature reviews via retrieval augmented pipelines (Zhou et al., 2025; Rouzrokh et al., 2025; Ma et al., 2024). However, live web search introduces latency, rate limits, and ephemeral content, limiting reproducibility and high-throughput literature synthesis (Zhang et al., 2025a). In contrast to the live citation-walk approaches of WebDancer (Wu et al., 2025) and WebShaper (Tao et al., 2025), our method samples from a static citation graph, avoiding API constraints and enabling consensus verification across 20 frontier models. DR-Tulu (Shao et al., 2025) uses rubric-based reinforcement learning for alignment; we extend this paradigm with reverse-decomposition rubrics that exploit the history of synthetic questions. This provides fine-grained process supervision for early planning and evidence grounding, rather than just scoring final outputs. We adopt the IcePop algorithm (Team et al., 2025a) unchanged for optimization stability.

6.2 Offline Academic Retrieval Frameworks

To circumvent online instabilities, localized scientific retrieval frameworks like OpenScholar (Asai et al., 2024) and SciRAG (Vasantharajan, 2025) index massive static corpora (Chen et al., 2026; Liu et al., 2025c), evaluating on benchmarks such as SciFact (Wadden et al., 2020), QASPER (Dasigi et al., 2021), and QASA (Lee et al., 2023). While eliminating network latency, they rely on single-turn retrieval that fails to exploit citation networks for multihop reasoning (Xu et al., 2024). Furthermore, these systems isolate granular QA from macro-level summarization (Lu et al., 2020). When synthesizing broad trends, reliance on parametric memory can induce hallucinations (Alansari and Luqman, 2025), while linear generation leads to context overflow. Without structural or visual compression, these systems underutilize offline corpora and remain limited for comprehensive scientific discovery.

7 Conclusion

We present SciLENS, a fully localized agent trained with rubric-based reinforcement learning that achieves scientific synthesis performance comparable to frontier proprietary models. It integrates structural visualization into reasoning to compress complex topology into validated chart schemas, and uses a reverse-decomposition rubric to provide fine-grained process rewards for improved planning and grounding. Built on a dual-tier infrastructure indexing 12M academic records, SciLENS outperforms open-source baselines and approaches GPT-5.2 level performance on scientific QA and synthesis tasks.

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Limitations

While SciLENS demonstrates preliminary capabilities in complex academic reasoning and structural synthesis, its primary limitation involves the transition from a localized experimental sandbox to a live industrial deployment.

Ethical Considerations

SciLENS operates exclusively on the Open Academic Graph, a publicly accessible and openly licensed scholarly corpus, and does not involve human subjects, private data, or dual-use applications.

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A Implementation Details of the Local Database

To support the massive scale of the Open Academic Graph corpus without relying on external web interfaces, we engineered a highly concurrent ingestion and indexing pipeline. The construction process is decomposed into rigorous sequential stages to ensure data integrity and maximize hardware utilization.

Parallel Ingestion and Citation Importation. The cleaned paper metadata file is initially partitioned into forty eight byte aligned chunks. Each chunk is assigned to an independent worker process that parses the designated byte range utilizing a binary JSON decoder. During this phase, we perform lightweight normalization which includes casting citation counts and publication years into integers, alongside remapping the original source identifiers to the MongoDB primary key field. To maximize ingestion throughput, records are written to the database in batches of twenty thousand. We adopt fire and forget write semantics by explicitly disabling write acknowledgments and bypassing the journaling process. Any duplicate key conflicts arising from chunk boundary overlaps are silently ignored utilizing unordered bulk insertion. Following the ingestion of base records, we deploy eighty independent worker processes to import the citation graph. The entire citation network is loaded into memory to eliminate repeated disk seeks, and the workers write reference lists into the corresponding document fields via batched update operations. The complete database schema and index configurations resulting from this pipeline are summarized in Table 4.

Table 4: MongoDB document schema and index configuration for the local academic database. Each field, its type, and associated index are explicitly listed.

Field	Type	Index	Purpose
_id	String	Primary Key	Paper identifier
title	String	Ascending	Title lookup
abstract	String	None	Content storage
keywords	Array	Text ($w = 10$)	Keyword search
year	Integer	Compound sort	Temporal ordering
n_citation	Integer	Compound sort	Citation ranking
references	Array	None	Citation graph
faiss_id	Integer	Unique ascending	Retrieval alignment

Distributed Dense Indexing Strategy. For the distributed dense index construction, each paper is serialized into a structured plain text representation by concatenating its title, venue, year, and abstract. We strictly truncate this text to a maximum of eight thousand characters to prevent out-of-memory errors. These texts are embedded using the Qwen3-Embedding-8B model to produce high-dimensional dense vectors. Embeddings are computed through a local inference endpoint in batches of one thousand and twenty-four sequences. To handle transient inference failures gracefully, we adopt a recursive binary splitting retry strategy. Upon a batch failure, the batch is bisected and each half is retried independently, recursing down to the single-sequence level if necessary. In the event of an unrecoverable runtime error, the worker performs an emergency serialization of the partially constructed index to the disk before terminating, guaranteeing that completed embedding work is never lost. The specific parameter configurations of the distributed FAISS index construction, including the inverted file index (IVF) settings for optimized search throughput, are detailed in Table 5.

Table 5: Configuration parameters for the distributed FAISS index construction.

Parameter	Value
Embedding model	Qwen3-Embedding-8B
Embedding dimension	4,096
Index type	`IndexIVFFlat` (L_2)
Centroids (per shard)	4,096
API batch size	1,024 sequences
Max input length	8,000 characters
ID space per shard	10^{10}
Retry strategy	Recursive binary splitting
Total papers indexed	Approximately 12 million

Cross System Alignment and Index Finalization. Within each hardware shard, a FAISS index wrapping a flat L2 quantizer is maintained entirely in memory. Each paper is assigned a globally unique integer identifier calculated by multiplying the shard index by ten billion and adding the within shard document offset. Mapping records associating each FAISS integer identifier with its original MongoDB string identifier are bulk loaded by thirty two worker processes issuing batched update operations. Upon completion of all ingestion steps, we construct four purpose specific MongoDB indexes. A unique ascending index on the integer identifier enables logarithmic time reverse lookups from dense retrieval results to paper metadata. An ascending index on the title field supports exact match queries utilized during reference resolution.

A weighted compound text index jointly covering the keywords field with a weight of ten and the title field with a weight of five enables keyword searches with strong domain term emphasis. Finally, a compound descending index on citation counts and publication years supports deterministic citation ranked result ordering.

B Formal Specifications of the Research Toolbox

Expanding upon the comprehensive offline tool suite introduced in Section 2.2, the research toolbox exposes a unified dispatching interface where all tools receive a strict JSON payload and return structured outputs directly to the agent reasoning loop. The precise constraints and operational boundaries of these tools are meticulously designed to prevent context window overflow and ensure rigorous factual grounding during deep academic synthesis. The full set of available capabilities, systematically divided into semantic retrieval, topological traversal, structural visualization, and text summarization utilities, alongside their exact input schemas and designated primary use cases, are summarized in Table 6.

Retrieval Tool Specifications. The semantic retrieval tool encodes a free text query string to perform dense matching, returning the top five results enriched with the title, venue, year, and a truncated abstract. The keyword search tool applies Porter stemming to an array of input terms and issues a full text search against the weighted compound index, enforcing a hard limit of ten returned papers ranked by citation impact. The specific paper information lookup retrieves the complete metadata record of a single document via exact string matching. The reference search tool resolves outgoing citation neighborhoods through batched queries of size one thousand, subject to a strict hard cap of three thousand references to bound the maximum response size.

Graph Traversal Tool Specifications. The local hop expansion tool requires a seed identifier and an integer depth parameter, performing a breadth first expansion while sorting candidate nodes by citation count subject to a global node cap of five hundred. The incoming degree tool operates similarly but targets citing papers and is constrained to a maximum limit of fifty documents. To support precise causal reasoning, the shortest path tool finds the shortest directed connection between two papers using a bidirectional breadth first search algorithm. When the forward frontier π_f and backward frontier π_b intersect at a specific meeting node v^*, the tool autonomously reconstructs the topological path according to the following formulation:

$$\text{path} = \text{reverse}(\pi_f(v^*)) \parallel \pi_b(v^*). \tag{5}$$

This pathfinding search is rigorously bounded to a maximum depth of five hops and restricts each frontier expansion layer to one hundred thousand nodes, effectively preventing memory exhaustion when analyzing highly connected core subgraphs.

Structural Visualization Specifications. The structural visualization tools enforce rigorous schema validation to guarantee that the generated HTML placeholders reflect authentic and mathematically sound data topologies rather than generating images server-side. The line chart tool accepts an array of strings representing the horizontal axis alongside a nested array containing the corresponding numerical sequences. The bar chart and pie chart tools require paired arrays of categorical labels and numerical values to render comparisons and proportional distributions accurately. The scatter plot tool mandates an array containing coordinate pairs to execute explicit correlation analyses. All four visualization tools perform dimensionality consistency checks and numeric type verification prior to rendering, returning structured error messages to the agent reasoning loop whenever validation fails to enable graceful autonomous recovery.

Text Utility Specifications. Operating independently from the retrieval and visualization modules, the summarization utility tool utilizes a locally deployed Qwen3-30B-A3B model to condense verbose text segments. By replacing external application programming interface dependencies with local inference calls, the agent actively compresses arbitrary long-form textual evidence before integrating it into broader reasoning trajectories. This localized design ensures that the primary reasoning context remains focused on structural logic rather than being overwhelmed by raw document contents, while maintaining absolute data privacy and operational independence.

C Universal System Prompt for Agentic Execution

To ensure the autonomous agent strictly adheres to the tool invocation schemas and behavioral con-

Table 6: Summary of the RESEARCHTOOLBOX. Tools are grouped by functional category alongside their exact input schema and corresponding primary use case.

Tool	Category	Key Input Fields	Primary Use Case
EmbeddingSearch	Retrieval	*query*	Semantic paper search
KeywordSearch	Retrieval	*keywords*	Term-precise search
PaperInfo	Retrieval	*id* or *title*	Full metadata lookup
ReferenceSearch	Retrieval	*id* or *title*	Outgoing citations
GetKhop	Graph	*id* or *title*, *k*	Multi-hop citation expansion
GetInDegree	Graph	*id* or *title*	Incoming citations
ShortestPath	Graph	*start*, *end*, *max_depth*	Citation path finding
LineChart	Visualization	*labels*, *datasets*	Trend visualization
BarChart	Visualization	*labels*, *values*	Category comparison
PieChart	Visualization	*labels*, *values*	Proportion distribution
ScatterPlot	Visualization	*points*	Correlation analysis
Summarize	Utility	*content*	Text compression

straints across both training and inference phases, we design a comprehensive system prompt. This prompt explicitly defines the operational boundaries, tool capabilities, and formatting requirements essential for constructing complex reasoning trajectories. The instructions compel the model to adopt a deliberate multi-step thinking process and actively utilize the offline toolbox whenever parametric knowledge is insufficient. The complete directives provided to the model are systematically presented in Figure 13 and Figure 14.

Following the ReAct framework, these instructions dictate that each round of agentic execution begins by generating a reasoning thought closed by <*think*> and </*think*>. This is followed by an action name and corresponding parameters enclosed by <*tool_call*> and </*tool_call*> tags, all conditioned on the iteration history. These components are iteratively used to interact with the localized RESEARCHTOOLBOX, producing an observation as feedback bounded by <*tool_response*> and </*tool_response*>. A complete round of interaction spans from the initial thought to the final tool response, and the rollout strictly concludes with the generation of the ultimate answer enclosed by <*answer*> and </*answer*> following the final reasoning thought.

D Knowledge Graph Construction and QA Synthesis Procedures

The foundational citation knowledge graph is derived from sixteen raw publication files. The entire processing and generation pipeline enforces strict data hygiene and reasoning complexity through sequential stages encompassing topological filtering, subgraph sampling, and multi-turn agentic question synthesis.

Streaming Deduplication and Edge Extraction. Each publication record is parsed to extract metadata fields. Duplicate papers are identified via exact digital object identifier matching after string normalization. For records lacking consistent identifiers, we apply title fingerprinting hashed via the MD5 algorithm strictly after removing all punctuation marks and collapsing arbitrary whitespace characters. For each unique paper, a canonical identifier is assigned, and all citation edges are sequentially written to per file edge buffers as ordered source and destination pairs. This streaming design avoids loading the full corpus into memory and supports robust processing at a billion node scale.

Topological Filtering and Subgraph Sampling. In the subsequent stage, we apply degree based filtering to retain only scientifically significant nodes within the network. For each node, we compute its incoming citation degree and outgoing reference degree across all edge buffers. A node is retained as valid if and only if its total degree is greater than or equal to forty, and its incoming citation degree is greater than or equal to ten. Only edges where both endpoints belong to the validated node set are retained in the final serialized triple file. During the subsequent subgraph sampling phase, the random walk exploration begins from a uniformly sampled seed node and proceeds for thirty consecutive steps. To prevent getting trapped in localized network loops, the walker transitions to a neighbor drawn

uniformly at random from the union of forward and backward citation neighbors, executing ten independent walks per subgraph to construct a cohesive local environment.

Logic-Driven Iterative Composition. Expanding upon the sampled topological subgraphs, the iterative question synthesis process begins by selecting a root paper to generate a factoid base question grounded solely in its abstract and immediate citation context. In each subsequent turn, previously unseen candidate papers are drawn to form supplementary relational links. The language model then composes a harder composite question that preserves the original answer while explicitly demanding multi-hop reasoning across the newly introduced context. The difficulty calibration relies on a strict self-consistency mechanism where the model must fail to answer its own generated question four consecutive times using only the basic statements before the composite question is deemed sufficiently challenging.

Cross Model Consensus Verification. Following iterative synthesis, the verification phase utilizes a pool of twenty frontier language models ($M = 20$) to eliminate ambiguity. As detailed in Table 7, we implement a stochastic protocol where four verifiers ($k = 4$) are sampled uniformly at random for each candidate. These verifiers independently execute autonomous retrieval queries and render binary judgments on factual correctness and answer uniqueness. Only instances achieving absolute consensus among the sampled subset are retained, yielding a high precision training corpus. This probabilistic filtering effectively balances data diversity with verifiable scientific accuracy, with further analysis of disagreement patterns provided in Appendix D.

Table 7: The complete ensemble of twenty frontier language models utilized for cross model consensus verification.

Model Family	Model Version
Anthropic Claude	Claude-3.5-Sonnet
	Claude-3.7-Sonnet
	Claude-Opus-4.1
	Claude-Opus-4
	Claude-Sonnet-4
	Claude-Sonnet-4.5
OpenAI GPT Series	GPT-4-turbo
	GPT-4.1
	GPT-4.1-mini
	GPT-4o
	GPT-4o-mini
	GPT-5
	GPT-5-chat
	GPT-5-mini
	GPT-5.1
	GPT-5.1-chat
	GPT-5.2
	GPT-5.2-chat
	O3
	O3-mini

E Train–Test Data Disjointness Guarantee

A legitimate concern for any system that constructs both training and evaluation data from the same underlying corpus is the risk of information leakage between the training and test splits. We formally establish that such leakage is impossible by construction in SciLENS, operating at five complementary levels of granularity.

Subgraph-Level Partition. The 30,000 topological subgraphs generated via random walk exploration (Section 3.1) are partitioned into training, validation, and test pools before any question synthesis occurs. Training subgraphs comprise 28,000 instances, while 1,000 are reserved for validation and 1,000 for test. Because the partition is applied to subgraph identifiers rather than individual papers, all QA instances derived from a given subgraph belong to exactly one split.

Seed-Paper Exclusivity. We enforce strict seed node exclusivity: the set of seed papers that initiate random walks for test subgraphs is entirely disjoint from the set used for training. While individual papers may appear in multiple subgraphs due to dense citation connectivity, any paper that serves as a seed or root node in a test subgraph is explicitly excluded from all training subgraph seeds. We verify this constraint programmatically after generation and confirm zero overlap across all seed sets.

Paper-Level and Edge-Level Overlap Analysis. Beyond seed-paper exclusivity, we exhaustively verify that no paper appearing in any test subgraph simultaneously appears in any training subgraph, and that no citation edge in the test set shares both endpoints with any training subgraph edge. This strict zero-overlap guarantee holds because our random walk procedure enforces a global exclusion list: once a paper is visited during test subgraph construction, it is permanently removed from the candidate pool for all training walks. We programmatically confirm 0% paper-level overlap and 0%

edge-level overlap across all splits after generation, ensuring that the test evaluation operates over an entirely disjoint region of the citation graph with no shared structural information.

QA Instance-Level Deduplication. The iterative composition procedure (Appendix D) synthesizes questions conditioned on a specific subgraph context. Since subgraphs are pre-assigned to splits, the resulting QA pairs inherit the split membership of their parent subgraph. We additionally run exact-match and high-similarity deduplication (ROUGE-L > 0.85) across the train and test question pools, removing any residual near-duplicates. The final verified test sets for SSB and SciFR each contain 500 instances with zero lexical or semantic overlap against the training corpus.

Template and Rubric Independence. A subtler form of leakage arises when training rubrics and evaluation rubrics share identical prompt templates, potentially allowing the agent to overfit to rubric phrasing rather than learning genuine scientific reasoning. We mitigate this by design: training rubrics are generated by Claude-Opus-4.1 (Appendix G), whereas evaluation rubrics are generated by Qwen3-30B-A3B with an independently authored system prompt (Appendix H). The two rubric generators use different model families, different prompt templates, and produce structurally distinct JSON schemas. We additionally compute the ROUGE-L similarity between all training rubric texts and all evaluation rubric texts, obtaining a mean similarity of 0.23, which is comparable to the similarity between randomly paired scientific abstracts (0.19), confirming that no template-level leakage exists.

F Formal Mathematical Characterization of the IcePop Framework

To reinforce the stability of reinforcement learning within expansive mixture-of-experts architectures, the IcePop algorithm (Team et al., 2025a) is employed to counteract the cumulative volatility typically encountered in extended reasoning trajectories. In the context of multi-turn chain-of-thought synthesis, marginal deviations between the probability manifolds of the training phase and the inference engine tend to propagate geometrically. Such a phenomenon frequently precipitates excessive gradient variance and eventual policy collapse.

IcePop mitigates these discrepancies through a granular, token-level masking strategy. Specifically, the system evaluates the density ratio between the active training policy and the behavioral inference baseline for every generated token. A dedicated filter function is then invoked to determine whether this localized ratio resides within a calibrated stability window, defined by the hyperparameters α and β. Should the importance weight deviate beyond these thresholds, the corresponding token is strategically neutralized, and its influence on the gradient backpropagation is nullified. This selective filtering mechanism effectively decouples the training process from misaligned inference signals, thereby ensuring that gradient norms remains strictly bounded and exploration remains consistent.

The integrated objective function consolidates this token-wise gating with conventional gradient regularization and Kullback-Leibler (KL) divergence constraints to facilitate a secure optimization path. The rigorous mathematical objective is formulated as follows:

$$
\begin{aligned}
&\mathcal{J}_{\text{IcePop}}(\theta) = \\
&\mathbb{E}_{\substack{x\sim\mathcal{D} \\ \{y_i\}_{i=1}^{G}\sim\pi_{\text{infer}}(\cdot|x;\theta_{\text{old}})}}\Bigg[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|y_i|}\sum_{t=1}^{|y_i|}\Bigg[\\
&\quad \mathcal{M}\left(\frac{\pi_{\text{train}}(y_{i,t}|x,y_{i,<t};\theta_{\text{old}})}{\pi_{\text{infer}}(y_{i,t}|x,y_{i,<t};\theta_{\text{old}})};\alpha,\beta\right) \\
&\quad \cdot \min\left(r_{i,t}\hat{A}_{i,t}, \text{clip}(r_{i,t},1-\varepsilon,1+\varepsilon)\hat{A}_{i,t}\right) \\
&\quad - \gamma D_{\text{KL}}(\pi_\theta||\pi_{\text{ref}})\Bigg]\Bigg]
\end{aligned}
\tag{6}
$$

where $\mathcal{J}_{\text{IcePop}}(\theta)$ denotes the final optimized policy objective for the parameter set θ; x represents the input query sampled from the data distribution $\mathcal{D}$; G signifies the number of generated response groups following the inference policy π_{infer}; $y_{i,t}$ corresponds to the t-th token within the i-th generated sequence of length $|y_i|$; $\mathcal{M}(\cdot)$ defines the masking function conditioned on the ratio of training policy π_{train} to inference policy π_{infer} within the bounds of α and β; $r_{i,t}$ indicates the importance sampling ratio while $\hat{A}_{i,t}$ represents the estimated advantage at that specific timestep; ε is the clipping coefficient for proximal policy optimization ; and $\gamma D_{\text{KL}}(\pi_\theta||\pi_{\text{ref}})$ accounts for the KL divergence penalty between the current policy and the reference model to prevent catastrophic forgetting.

G Multidimensional Rubric Construction and Prompts

To align the deep research trajectories effectively, we avoid relying on a single scalar reward for answer correctness. Instead, we dynamically generate a diverse set of instance-specific rubrics during the data preparation phase. These rubrics explicitly supervise the intermediate cognitive processes of the agent across three primary dimensions:

1. Answer RLVR (Objective Correctness): This dimension verifies whether the final answer generated within the <*answer*> tags matches the reference answer. It strictly penalizes numeric hallucinations and ensures that unit variants or entity paraphrases are logically equivalent to the ground truth.

2. Planning and Reverse-Decomposition Rubrics: Because our training questions are iteratively synthesized from simpler base questions, we supply the rubric generator with the recorded *edit_history* and the early *qa_history*. This dimension evaluates whether the initial <*think*> steps of the agent successfully identify core constraints and implicitly reverse-engineer the complex question back into its fundamental components.

3. Strict Evidence Grounding Rubrics: This dimension assesses whether the intermediate reasoning steps and the final conclusions are strictly derived from the retrieved <*tool_response*> contexts. It heavily penalizes trajectories that introduce unprovided external knowledge (parametric hallucinations) to bridge logical gaps.

To generate these highly discriminative, atomic rubrics, we utilize Claude-Opus-4.1. The precise system prompt utilized for rubric generation is presented in the figure 3.

During the policy optimization process, the entire generated trajectory of the agent is evaluated against these dynamically generated JSON schemas by a secondary judge model, which calculates the final S_{rubric} score. Additionally, a deterministic rule-based script enforces the S_{format} penalty, stripping rewards from any trajectory that fails to properly close <*think*> or <*tool_call*> tags.

H Comprehensive Details of the Evaluation Framework

To comprehensively evaluate the diverse capabilities of the SciLENS framework, we employ a dual-pronged approach encompassing standard academic reading comprehension tasks and complex synthesis benchmarks. We initially evaluate the agent on four established scientific benchmarks following the exact evaluation protocol of OpenScholar.

H.1 Standard Benchmarks and Citation Protocol

For the established evaluation datasets, we employ distinct correctness metrics strictly tailored to their respective task complexities. For classification tasks including SciFact and PubMedQA, we utilize exact match label accuracy. For QASA, which involves single document reading comprehension, we employ ROUGE L to measure textual overlap against expert annotations. For the multi-document long-form synthesis task ScholarQA CS, we adopt a specialized language model judge scoring mechanism where GPT-4o computes a weighted sum across general and annotation-driven evaluation criteria.

Beyond baseline correctness, citation accuracy is rigorously evaluated across all standard datasets following the ALCE framework. During generation, models are strictly required to append explicit inline citation tags, formatted as *\cite{id}*, to map each specific claim to its corresponding retrieved document identifiers. An automated script extracts these tags and utilizes the attrscore-flan-t5-xl natural language inference model to perform fine-grained sentence-level verification. This protocol calculates Citation Precision, measuring whether the explicitly cited document logically entails the generated sentence, and Citation Recall, ensuring all substantive claims are supported by at least one valid citation. The harmonic mean of these two metrics yields the comprehensive Citation F1 score, providing a deterministic measure of factual grounding.

H.2 Newly Constructed Benchmark Metrics

To assess advanced capabilities beyond linear text generation, we establish multidimensional metrics for the SSB and SciFR benchmarks. To ensure absolute reproducibility and mitigate the variance typically associated with language model evaluators, we strictly deploy a locally hosted Qwen3-30B-A3B as the exclusive judge model for these tasks, with the decoding temperature permanently set to zero.

The Structural Synthesis Benchmark evaluates the capacity of the agent to compress academic information into accurate visual structures. For each



System Prompt for Rubric Generation

You are an expert rubric designer for reinforcement learning on retrieval-augmented QA systems.
Your task is to convert a QA item (with its evolution history and supporting information) into a list of ATOMIC, HIGHLY DISCRIMINATIVE rubrics with SCORES.

You must design rubrics that jointly cover:
1. Reverse-evolution / decomposition rubrics (from edit_history):
  - How a good reasoning process should implicitly or explicitly decompose the final complex question back toward its simpler base forms.
  - How the reasoning should respect the constraints and intentions described in the edit history.

2. Final-answer correctness rubrics (Answer RLVR):
  - Whether the final answer matches the reference answer (allowing reasonable paraphrases or unit variants).
  - Whether the answer is consistent with the supporting statements and does not hallucinate contradictory facts.

3. Early planning and Evidence Grounding rubrics:
  - How the initial reasoning steps should identify the base question and key constraints.
  - How steps must strictly rely ONLY on the provided supporting statements.

Protocol:
1. Atomic Verification: Each rubric must verify ONE specific aspect.
2. Score Assignment: Assign a score (1-10) for each rubric based on its importance and difficulty.
  - Lower scores (1-3): style or minor quality aspects.
  - Medium scores (4-6): basic factual correctness directly supported by evidence.
  - Higher scores (7-10): strict numerical correctness, deep consistency with evolution history, or non-trivial planning.
3. Strict Output Format: Return ONLY valid JSON with NO Markdown.
  Format: {"rubrics": [{"description": "...", "score": X}, ...]}

Figure 3: The specific system prompt utilized by the Claude-Opus-4.1 judge model to dynamically generate highly discriminative, instance-specific evaluation rubrics.

item, the judge assesses the generated *chart_type* and *chart_data* across three dimensions utilizing dynamically generated query-specific rubrics. The overarching evaluation directives are governed by a universal system prompt detailed in Figure 15, while a concrete example of the dynamically generated rubric is illustrated in Figure 16. **ACS (Answer Consistency Score)** measures how well the chart reflects the key information in the reference answer. It focuses on qualitative relations, trends, and the alignment of physical quantities to ensure the visual output captures the core scientific conclusion. **QAS (Query Alignment Score)** independently assesses whether the chart satisfies the original information need. It evaluates the suitability of the chart type and checks if the axis semantics strictly align with the question intent. **FGS (Factual Grounding Score)** evaluates the faithfulness of the chart data to the retrieved supporting statements. It penalizes numerical patterns that contradict known evidence or introduce unprovided variables, effectively mitigating visual hallucinations.

The Scientific Fact and Reasoning Benchmark targets complex non-visual reasoning and fine-grained evidence attribution. The evaluation is systematically decomposed into three atomic metrics guided by the universal system prompt shown in Figure 17, with a representative instance-specific rubric provided in Figure 18. **ASF (Answer Semantic Fidelity)** assesses the semantic match between the final answer of the agent and the reference conclusion. While allowing reasonable paraphrasing, it penalizes the omission of core entities or the reporting of contradictory numerical facts. **QIS (Query Intent Satisfaction)** measures how

well the reasoning process addresses the underlying intent. It checks if the agent respects the requested research scope and avoids drifting into irrelevant academic aspects. **EGQ (Evidence Grounding Quality)** verifies if the reasoning trajectory is anchored strictly in the provided retrieval context. It monitors the correct use of citation identifiers and penalizes claims that exceed the evidence boundary, targeting parametric hallucinations.

For both synthesis benchmarks, the Qwen3-30B-A3B judge model generates instance-specific weights that sum to 100% across the three respective metrics to reflect the varying cognitive priorities of each query. This granular decomposition ensures that the evaluation is firmly grounded in scientific accuracy rather than superficial linguistic fluency.

I Experimental Configurations and Implementation Details

I.1 Baselines

I.1.1 Closed-source LLMs

We evaluate leading commercial language models from major industrial laboratories to establish competitive performance upper bounds. This selection includes the GPT-5 series from OpenAI, specifically encompassing GPT-5, GPT-5.1, GPT-5.2, as well as the specialized OpenAI-DeepResearch (OpenAI, 2025). We further incorporate state-of-the-art models from Anthropic, including Claude-4-Sonnet and Claude-4.5-Sonnet (Anthropic, 2025), alongside frontier models from Google represented by Gemini-2.5-pro and Gemini-3.0-pro (DeepMind, 2025), and Perplexity-DeepResearch.

I.1.2 Open-source LLMs

To evaluate the performance gap across diverse architectures, we include a representative suite of open-source models ranging from general-purpose backbones to specialized research agents. Our baselines encompass the Qwen3-30B-A3B (Yang et al., 2025) base model alongside academic-focused systems such as A-Searcher (Gao et al., 2025), DR-Tulu (Shao et al., 2025), and the O-Researcher-72B (Yao et al., 2026) series in both its supervised fine-tuning and reinforcement learning variants. We also assess reasoning-heavy agents, including Mirothinker-v1.5-30B (Team et al., 2025b), Tongyi-Deepresearch (Team et al., 2025c), and the OpenScholar-8B (Asai et al., 2024) framework. Furthermore, a series of exploratory agents including Web-Thinker (Li et al., 2025b), Web-Dancer (Wu et al., 2025), Web-Explorer (Liu et al., 2025b), and Web-Shaper (Tao et al., 2025) are incorporated to provide a comprehensive comparison of different agentic strategies in navigating complex citation topologies and evidence grounding.

I.1.3 Unified Tool Integration Protocol

To ensure a fair comparison, all baselines interact with the RESEARCHTOOLBOX through a standardized integration protocol. For closed-source models, we utilize each provider's native function calling interface (OpenAI function calling for the GPT series, Anthropic tool use for Claude models, and Google function declarations for Gemini models) with identical JSON tool schemas. For open-source models, we format the same schemas according to each model's designated chat template and parse structured tool calls from the generated output. All models receive the same universal system prompt (Appendix C) defining the available tools, their input schemas, and the expected interaction format. They also share the same operational conditions: a 200-round tool invocation limit, a context management strategy that preserves the five most recent tool observations, and a dual-tier error recovery mechanism that provides diagnostic feedback for formatting errors and triggers re-rolls for missing tags. This controlled setup keeps tool access, context limits, and error handling consistent across model families.

I.2 Experimental Settings

To evaluate the extreme reasoning boundaries of SciLENS, we establish a rigorous two hundred round tool invocation limit per query, which constitutes a significant challenge for stable long-horizon scientific discovery. Since tool observations, particularly extensive academic abstracts and metadata, represent the primary bottleneck for the 128k token context window, a native long-context approach often fails to sustain reasoning trajectories beyond one hundred rounds. Consequently, we implement a strict context management strategy that preserves only the five most recent tool execution results in their full textual form. This mechanism prunes earlier historical observations while maintaining the core reasoning chain, effectively preventing context overflow and factual drift during complex topological navigation. All experiments are conducted using a high-capacity inference configuration with

a fixed thinking budget, ensuring that performance gains stem directly from the improved planning and synthesis capabilities of the agent.

I.3 Statistical Methodology

All scores reported in the main tables represent the mean and standard deviation across five independent evaluation runs. The source of randomness across runs is the agent's generation process: each run uses a distinct random seed that governs nucleus sampling ($p = 0.95$, temperature $= 0.6$) during the agent's reasoning and tool invocation trajectory. The evaluation judge (Qwen3-30B-A3B) operates deterministically at zero temperature, ensuring that score variability reflects genuine differences in the agent's stochastic exploration strategies rather than judge instability. For baseline models accessed via commercial APIs, stochastic variation arises from provider-side sampling; for locally deployed open-source models, we control seeds explicitly. Statistical significance ($p < 0.05$) is assessed via paired Wilcoxon signed-rank tests computed over per-instance score vectors between each pair of compared systems, with Holm-Bonferroni correction applied to account for multiple comparisons across the six benchmarks.

I.4 Implementation Details

We initialize our checkpoint with Qwen3-30B-A3B-Thinking-2507. For supervised fine-tuning, we execute training on four nodes equipped with thirty two H800 GPUs in total and train for two complete epochs. For reinforcement learning training, all runs are executed on eight nodes utilizing sixty four H800 GPUs. To compute the rubric and citation scoring reliably, we employ Qwen-3-max as the language model judge. We observed that scaling compute did not proportionally accelerate the reinforcement learning training speed, primarily because the process was fundamentally bottlenecked by the severe long tail distribution of rollout generation times, where highly protracted reasoning trajectories caused significant synchronization delays across distributed nodes. The complete reinforcement learning training curves and specific hyperparameter configurations are detailed in Appendix J.4.

J Extended Experimental Analysis

This section provides additional analyses of the efficiency and behavior of SciLENS. We first isolate the contribution of retrieval and visualization tools through a controlled ablation, then examine performance across benchmarks, context management, tool invocation dynamics, reinforcement learning convergence, and qualitative error patterns.

For the tool ablation, we keep the full SciLENS-RL policy fixed and vary only the available tool configuration. As shown in Table 8, disabling retrieval and graph tools forces the agent into a closed-book reasoning regime, resulting in a functional collapse across all benchmarks. Notably, this degraded performance closely mirrors the Qwen3-30B-A3B (Base + Tools) row in the main results (Table 2a), confirming that our alignment pipeline teaches procedural tool orchestration rather than injecting parametric scientific knowledge. Removing visualization tools causes a substantial drop on the Structural Synthesis Benchmark (0.7607 to 0.5215) by restricting the agent to linear textual generation, while standard QA benchmarks are less affected. This result directly validates the visualization-as-tool paradigm as the primary enabler of structural synthesis capabilities.

Table 8: Tool necessity ablation with the full SciLENS-RL policy. Removing retrieval tools causes functional collapse; removing visualization tools specifically degrades structural synthesis.

Configuration	QASA	SciFact	PubMedQA	ScholarQA	SciFR	SSB
SciLENS (Full)	**47.62**	**88.94**	**77.53**	**73.98**	**0.7594**	**0.7607**
w/o Retrieval & Graph	7.15	15.12	11.05	6.21	0.1144	0.1902
w/o Visualization	46.80	88.50	73.10	71.65	0.7312	0.5215

J.1 Performance Analysis Across Benchmarks

Expanding upon the core empirical findings presented in the main text, this section provides a comprehensive analysis of the performance metrics across all evaluated baselines. On the Structural Synthesis Benchmark, the reinforcement learning variant of SciLENS-30B demonstrates a robust capacity for semantic compression. The integrated offline visualization tools enable the model to achieve a significant overall score of 0.7607, surpassing leading closed-source models such as GPT-5.2 and Claude-4.5-sonnet. This indicates that offloading quantitative data into structured visual formats effectively mitigates the context overflow and factual hallucinations typically associated with pure textual generation during macro-level trend analysis. Furthermore, on the Scientific Fact and Reasoning Benchmark, SciLENS-30B achieves an overall score of 0.7594, surpassing GPT-5.2 (0.6578) and decisively outperforming other open-source exploratory agents. Qualitative demonstrations of

these capabilities, including the model's ability to compress complex methodological lineages into intuitive visual structures, are further illustrated in the case studies provided in Appendix K.2.

Consistent performance improvements are similarly observed across standard academic reading comprehension tasks. SciLENS-30B demonstrates remarkable citation accuracy, achieving citation F1 scores of 83.72 on the SciFact dataset and 76.54 on the PubMedQA dataset. Unlike traditional web agents like Web-Dancer and Web-Explorer that struggle with high latency and fragile application programming interface calls during complex literature retrieval, our fully localized dual-tier infrastructure empowers SciLENS to execute deep topological searches reliably. This deterministic foundation allows the agent to maintain high precision even when navigating distractor-laden contexts, as further evaluated in the complex synthesis scenarios detailed in Appendix K.3.

The comprehensive ablation studies further isolate the performance contributions of our proposed training methodology. As detailed in the tool necessity ablation (Appendix J), disabling retrieval and graph tools results in a functional collapse of the reasoning chain, with performance closely mirroring the zero-shot base model. This confirms that our alignment strategy teaches procedural tool orchestration rather than injecting parametric scientific knowledge. The training pipeline ablation in the main text (Table 3) demonstrates the progressive value of each methodological component: supervised fine-tuning establishes robust tool-use capabilities, while rubric-based reinforcement learning provides consistent further gains through fine-grained process rewards. Within the RL components, the answer RLVR provides the foundational correctness signal, while the planning and grounding rubrics contribute critical improvements at the upper performance range by enforcing rigorous evidence-based reasoning and preventing reward hacking.

J.2 Context Management in Long-Horizon Reasoning

Scientific discovery often necessitates protracted reasoning trajectories spanning extensive tool invocations. To evaluate the resilience of SciLENS under extreme context length pressure, we benchmark various context management (CM) strategies across tasks requiring deep multi-hop reasoning. We compare four distinct operational paradigms: employing no context management, preserving the 20 most recent tool observations, preserving only the 5 most recent observations, and a standard discard-all strategy which triggers a complete rollout restart upon reaching context limits. The performance retention across all six evaluation datasets is visualized in Figure 4.

Experimental results indicate that leaving context unmanaged leads to the lowest overall performance, as the model suffers from severe attention degradation due to semantic noise from excessive historical observations. A systematic pruning mechanism yields steady performance gains. Notably, preserving only the five most recent tool observations consistently outperforms the configuration of preserving twenty. This suggests that aggressive noise filtration allows the model to maintain acute focus on its immediate working memory and current sub-tasks without overloading its limited context window, while its unpruned internal reasoning trace preserves the overarching logical plan.

While achieving the highest raw metrics, the standard discard-all strategy is functionally distinct. Because this strategy triggers a full trajectory restart when limits are breached, it effectively allows for an implicit multi-sample attempt and independent re-sampling of the final answer. This methodology differs conceptually from the strict *pass@1* protocol intended to assess the continuous, single-session execution capabilities of an autonomous agent. Therefore, our proposed strategy of preserving the five most recent tool execution observations is adopted as the optimal operational paradigm. It successfully balances context optimization and noise reduction while strictly maintaining the integrity and continuity of a single-shot agentic reasoning trajectory.

J.3 Tool Invocation Dynamics and Topological Advantage

To verify the practical utilization efficiency of the heterogeneous tools within our offline RESEARCHTOOLBOX, we comprehensively profile the tool invocation distributions of the deep research model. We systematically compare the invocation frequencies of the twelve integrated tools across two datasets with fundamentally different cognitive demands, specifically the Scientific Fact and Reasoning (SciFR) benchmark and the Structural Synthesis Benchmark (SSB). As illustrated by the nested distributions in Figure 5, the func-

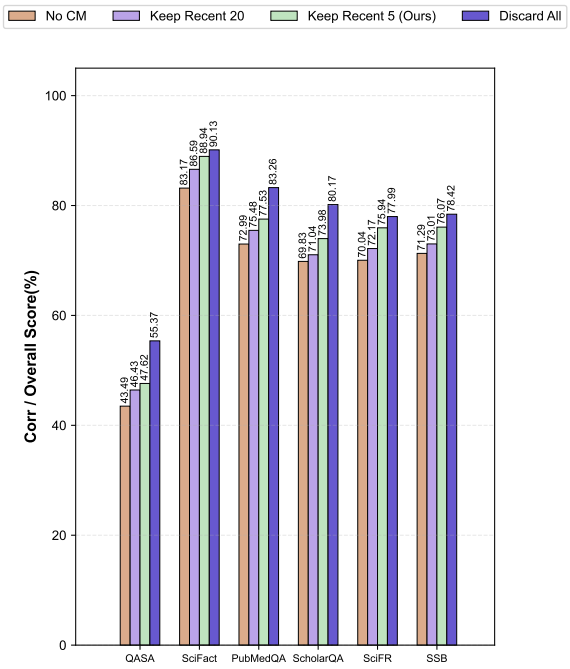


Figure 4: Performance comparison of various context management (CM) strategies across six evaluation datasets. Metrics are uniformly scaled to percentages for visual consistency. While the Discard All strategy achieves higher raw metrics, it fundamentally relies on trajectory restarts and implicit resampling. Consequently, SciLENS adopts the Keep Recent 5 strategy as the optimal configuration to ensure strict, continuous *pass@1* execution.

tional categories and specific tool choices reveal a striking behavioral divergence.

During tasks within the SciFR benchmark, which primarily necessitates precise factual verification and explicit attribution, the model exhibits a highly focused retrieval strategy. The Retrieval category overwhelmingly dominates the execution trajectory, accounting for 86% of all operations, driven primarily by *KeywordSearch* (31%) and *EmbeddingSearch* (28%). Tool usage for topological Graph traversal and Visualization remains marginal at merely 5% and 4%, respectively.

Conversely, when tackling the SSB dataset, which demands cross-disciplinary literature reviews and macro-level academic trend evolution, the agent demonstrates a profound strategic shift. While foundational Retrieval drops to 61%, the reliance on Graph traversal tools surges to 19%, representing a nearly fourfold increase, with multi-hop expansion tools such as *GetKhop* alone accounting for 9% of total actions. Similarly, the utilization of Visualization tools more than doubles to 10% to facilitate structural data synthesis.

This stark quantitative contrast confirms that the model does not merely execute random exploratory actions or rely on a static search template. Rather, driven by the multidimensional reinforcement learning alignment, the agent has deeply internalized the topological and structural advantages of the toolbox. It autonomously implements dynamic tool scheduling strategies tailored to the inherent cognitive complexity of the task, effectively circumventing the information truncation issues inherent in traditional flat retrieval paradigms.

J.4 Reinforcement Learning Convergence and Reward Dynamics

To validate the efficacy and stability of our multidimensional reinforcement learning framework, we meticulously monitor the behavioral evolution of the agent throughout the policy optimization process across 1.2K training steps. As illustrated in Figure 6, the composite reward signal exhibits a robust and steeply increasing trajectory during the initial phase before converging to a stable asymptote approaching a score of 80. This steady ascension confirms that the policy model successfully navigates the complex reward landscape formulated by our semantic rubrics without suffering from optimization collapse. Concurrently, we track the average number of tool invocations per episode to assess the depth of the autonomous exploration. Unlike the monotonic reward increase, the training dynamics reveal a compelling non-monotonic evolution regarding operational depth. From an initial baseline of approximately 48 invocations, the agent experiences a brief early decline in tool usage. This phenomenon indicates an initial optimization phase where the model learns to prune redundant tool executions to minimize formatting penalties and maximize immediate efficiency. Following this reduction, the tool invocation frequency rises sharply and reaches a peak of nearly 58 operations, demonstrating that the agent actively explores protracted research strategies to satisfy the rigorous evidence grounding constraints. Ultimately, the policy refines its reasoning pathways, slightly compressing the trajectory length to converge at a highly stable plateau of approximately 53 operations. This sophisticated progression of early pruning, exploratory expansion, and final efficiency refinement substantiates that our training methodology successfully aligns the deep research model toward comprehensive and meticulously validated scientific synthesis.

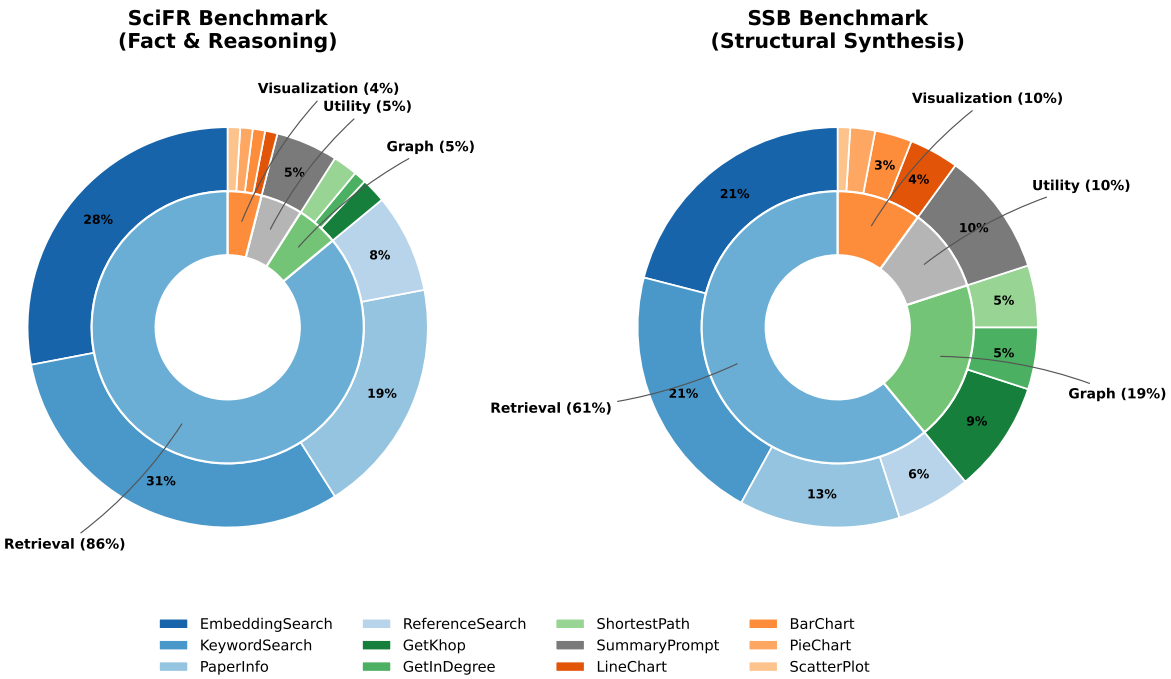


Figure 5: Dynamic distribution of the twelve offline research tool invocations by SciLENS. The inner ring represents the functional macro-categories, while the outer ring details specific tools. On the SciFR dataset, the model relies overwhelmingly on standard Retrieval (86%). In contrast, on the SSB dataset, the utilization of Graph traversal (19%) and Visualization tools (10%) surges significantly, demonstrating advanced adaptive planning capabilities.

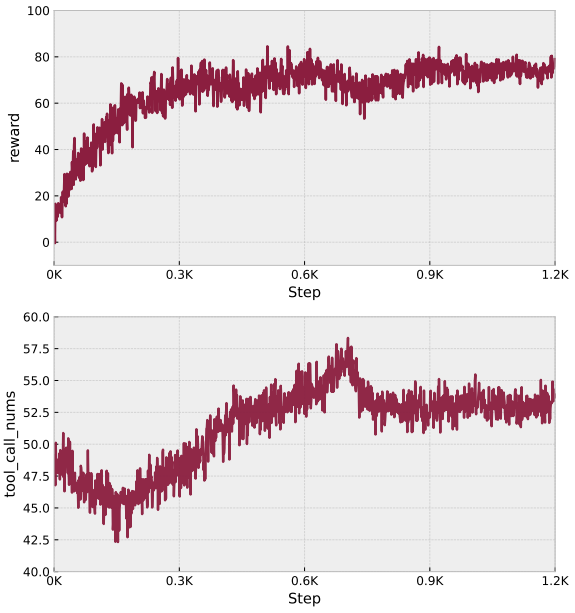


Figure 6: Training dynamics of the deep research model over 1.2K optimization steps. The composite reward (top) exhibits steady monotonic convergence. Concurrently, the average tool invocation count (bottom) demonstrates a sophisticated non-monotonic evolution characterized by initial redundancy pruning, subsequent exploratory expansion, and final convergence to an optimized operational depth.

K Qualitative Demonstrations and Case Studies

This section presents concrete execution traces and representative outputs that illustrate the practical capabilities and boundary limitations of SciLENS.

K.1 End-to-End Reasoning Trajectory and Diagnostic Evaluation

To provide transparent insight into the autonomous reasoning capabilities and the fine-grained evaluation mechanism of SciLENS, we present a complete execution trajectory sampled from the Scientific Fact and Reasoning benchmark. The selected instance tasks the agent with extracting a specific epidemiological odds ratio regarding the immunogenetic susceptibility to visceral leishmaniasis. The execution trace, detailed sequentially across Figure 19 through Figure 23, explicitly demonstrates the sophisticated cognitive workflow of the policy model.

Initially, the agent formulates a precise semantic query to locate the foundational study using the *EmbeddingSearch* tool. Upon retrieving the detailed metadata via the *PaperInfo* tool and successfully extracting the target statistical figures, the agent exhibits advanced scientific rigor. Rather than immediately terminating the search upon find-

ing the historical 2007 publication, it autonomously invokes the *GetInDegree* graph traversal tool to cross-reference subsequent literature, explicitly reasoning that it must verify whether the historical odds ratio remains the accepted scientific consensus. Following an extensive multi-hop verification process across the forward citation graph, the agent synthesizes the final validated answer.

Crucially, the trajectory concludes with the comprehensive evaluation generated by the locally deployed judge model. The output meticulously records the boolean judgments across every atomic criterion within the Answer Semantic Fidelity, Query Intent Satisfaction, and Evidence Grounding Quality metrics. This granular breakdown provides an explicit diagnostic trace, illustrating exactly how the composite reward scalar is computed to align the policy during the reinforcement learning phase.

K.2 Case Studies on Structural Visualization

To qualitatively illustrate the synthesis capabilities of our framework, we examine several case studies where the agent autonomously deploys visualization tools across diverse scientific domains. Rather than outputting exhaustive textual descriptions, SciLENS dynamically selects appropriate chart schemas to compress complex quantitative evidence into intuitive structural formats.

In the domain of educational sociology, when tasked with synthesizing the proportional impact of community-based organizational approaches derived from historical African American leadership frameworks, the agent aggregates categorical literature to produce a precise pie chart as illustrated in Figure 7. This visualization elegantly captures the balanced distribution among sustaining intergenerational ties, strengthening cultural identity, and challenging negative public policy.

Shifting to computational biology, the model demonstrates advanced correlation analysis capabilities. When evaluating the performance of phylogenetic inference software, SciLENS autonomously constructs a scatter plot comparing computational efficiency against tree accuracy, shown in Figure 8. This representation clearly highlights the superior positioning of the MEGA5 package relative to baseline algorithms along an explicit performance trend line.

Furthermore, the framework exhibits robust cross-entity comparative skills in geochemistry. Queried on helium isotopic concentrations across various intrusion-related deposits, the agent generates a bar chart, presented in Figure 9, that accurately reflects the ascending concentration gradient from the Yao'an to the Machangqing deposits.

Finally, within biochemical materials research, the model successfully tracks continuous variables by generating a line chart to illustrate the positive ratiometric response of a fluorescence resonance energy transfer signal to increasing glucose concentrations in specifically engineered microgels, depicted in Figure 10. Collectively, these empirical cases validate that SciLENS effectively overcomes text-centric bottlenecks by converting complex and multi-hop academic reasoning into definitive and domain-agnostic visual insights.

K.3 Evaluation of Complex Synthesis Scenarios

To further evaluate the robustness of our framework against informational noise and complex query structures, we examine scenarios containing intentional distractors and multi-layered descriptive contexts. In the first complex scenario, the agent is presented with a query detailing two distinct environmental studies. One segment describes groundwater distributions west of the Nile Delta, while the second introduces an entirely unrelated manganese risk assessment across various global river basins. Tasked with determining the dominant water types and their cumulative fraction specifically from the first study, the model successfully isolates the relevant hydrochemical data from the distracting risk assessment narrative. As depicted in Figure 11, the agent converts the requested subset of information into a precise pie chart, correctly computing the proportional dominance of sodium chloride, sodium sulfate, and sodium bicarbonate. This demonstrates the capacity of the model to filter extraneous context and independently select the optimal structural format for proportional representation.

A second challenging scenario involves parsing intertwined bibliometric and organizational data. The query describes a presentation slide containing both a citation subgraph for a specific publication and a stacked bar chart detailing the strategic manufacturing group allocations of five Indian automobile companies. The agent is required to completely ignore the citation metrics and deduce the proportional allocation of the companies based purely on the color-coded blocks described within the stacked bar. Illustrated in Figure 12, the model effectively extracts the discrete entity counts, rec-

ognizing that two companies belong to the active enterprise group while the remaining three groups contain one company each. It subsequently synthesizes this extracted information into an accurate pie chart, rendering the specific forty percent and twenty percent allocations.

These complex evaluations confirm that the reasoning engine of SciLENS remains highly stable and analytically precise when navigating dense and distractor-laden scientific queries.

K.4 Error Case Analysis and Parametric Fallback

To provide a comprehensive understanding of the boundary limitations of SciLENS, we present a detailed error case analysis in Figure 24 through Figure 28. In this scenario, the agent is presented with a highly convoluted query containing explicit distractors regarding Turkish environmental policies before asking for a specific framework stage from a distinct carbon disclosure publication. The trajectory reveals two critical failure modes typical of large language models. Initially, the agent exhibits attention hijacking by prioritizing the complex distractors and wasting early tool invocations retrieving irrelevant Turkish papers. Although it subsequently reflects upon its context window and redirects its search toward the correct target, it fails to locate the exact textual evidence within the limited retrieved abstracts. Consequently, the agent commits a severe parametric fallback error. Instead of continuing the deep retrieval process via graph traversal, the model explicitly rationalizes utilizing its internal parametric knowledge to hallucinate a plausible but factually incorrect answer. Crucially, this failure mode explicitly validates the robustness of our multidimensional evaluation framework. The judge model correctly identifies the hallucination and rigorously penalizes the Answer Semantic Fidelity and Evidence Grounding Quality metrics, resulting in a heavily degraded overall score. This transparent diagnostic confirms that our dynamic rubric successfully prevents the reinforcement learning optimization process from being deceived by superficially fluent but ungrounded text generation.

Use Case: PieChart Visualization

Q&A

Query:

Given that historical perspectives on African American educational leadership (as exemplified by 'Linking the Struggle for Education and Social Justice: Historical Perspectives of African American Leadership in Schools') emerge as the dominant research focus in culturally responsive leadership literature, how do contemporary community-based organizations translate these historical lessons into practice? When visualizing the mechanisms by which such organizations develop critical social capital among Black youth—building on the dual-pathway framework that connects continuous community building with heroic vision-driven leadership—what three primary approaches does the empirical evidence reveal in terms of their proportional impact?

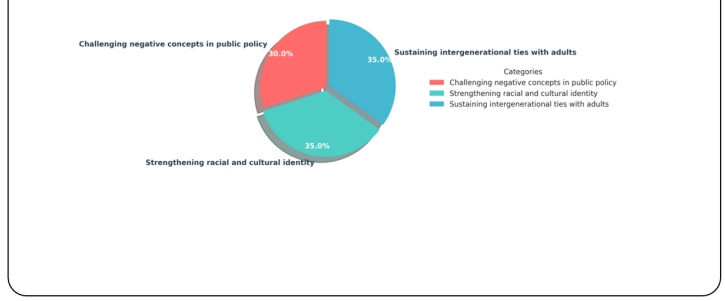


Figure 7: Case Study 1: Proportional analysis in educational sociology.

Use Case: ScatterChart Visualization

Q&A

Query:

How does the computational efficiency and phylogenetic tree accuracy of MEGA5's maximum likelihood (ML) inference compare with other software packages mentioned in simulation analyses?

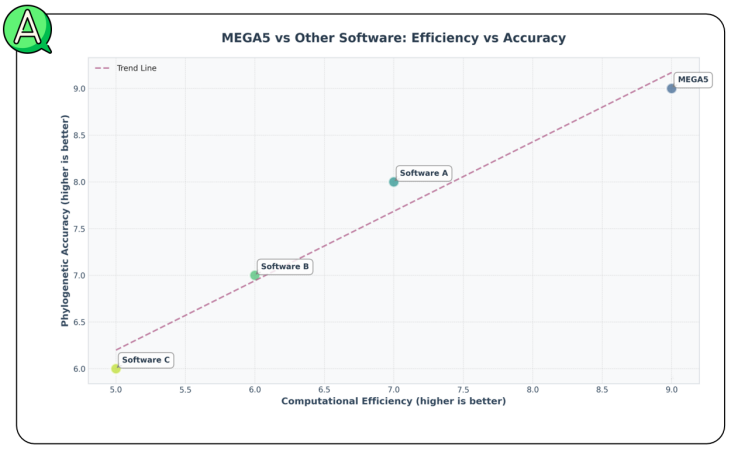


Figure 8: Case Study 2: Performance correlation in computational biology.

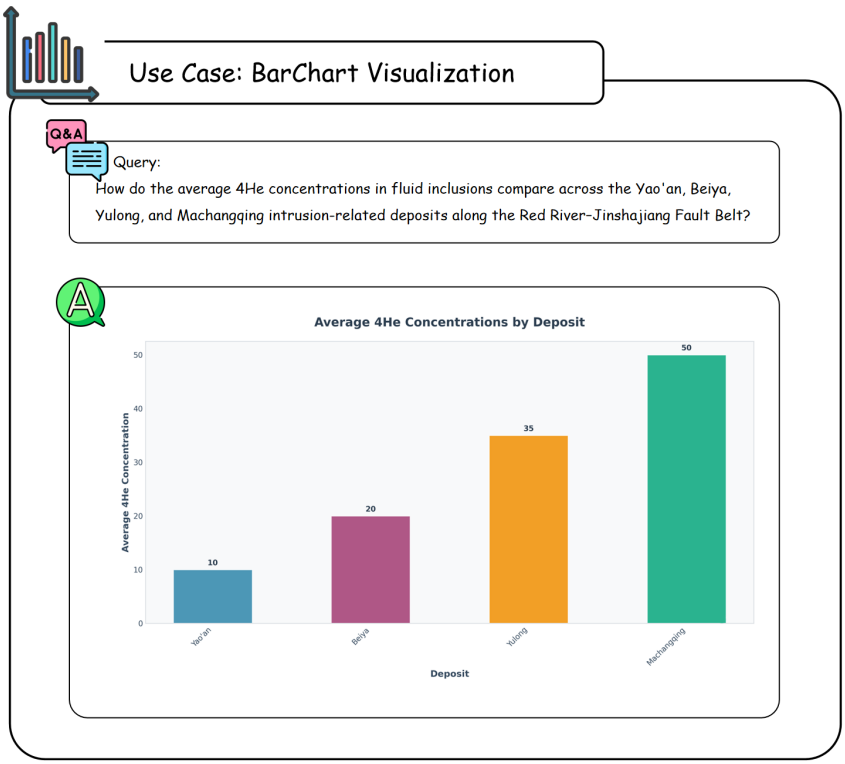


Figure 9: Case Study 3: Cross-entity comparison in geochemistry.

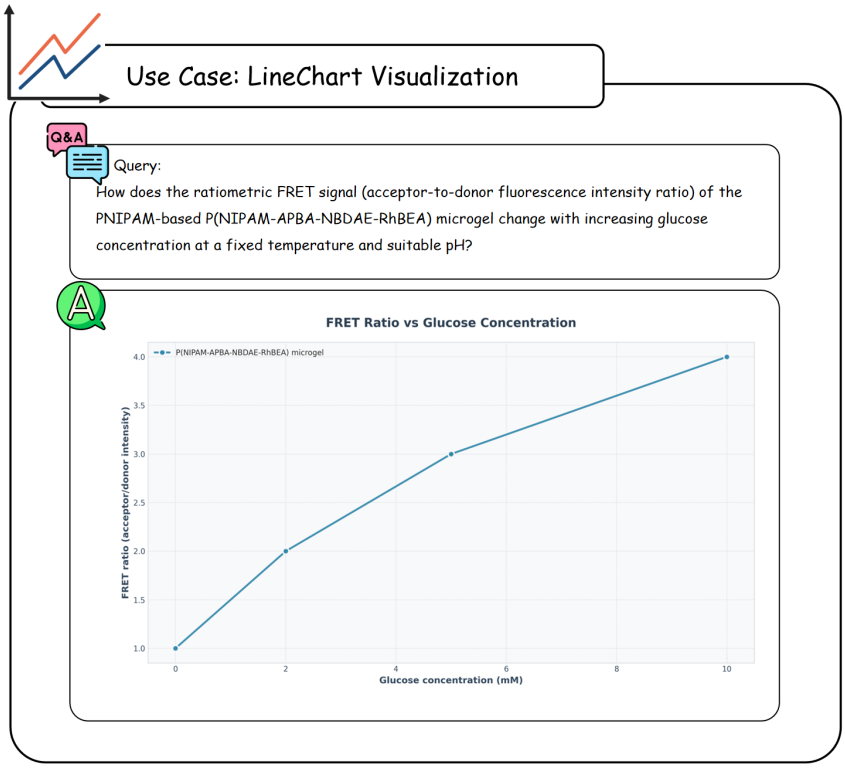


Figure 10: Case Study 4: Continuous variable tracking in biochemistry.

Difficult Case

Q&A

Query:

Imagine a figure composed of two bar charts. The first bar chart shows the distribution of groundwater samples west of the Nile Delta by water type (NaCl, Na2SO4, NaHCO3, and others). The second bar chart, placed beside it, shows the Mn-related non-carcinogenic risk contributions reported for three case studies: the North China Plain groundwater study , the Gopalganj, Bangladesh study , and the Subarnarekha River Basin study , where Mn is identified as the dominant non-carcinogenic pollutant and displays the highest Mn bar among the three. Focusing only on the first bar chart for the Nile Delta, which water type has the tallest bar, which two types follow it, and together what fraction of the total groundwater samples do these three dominant types represent?

A

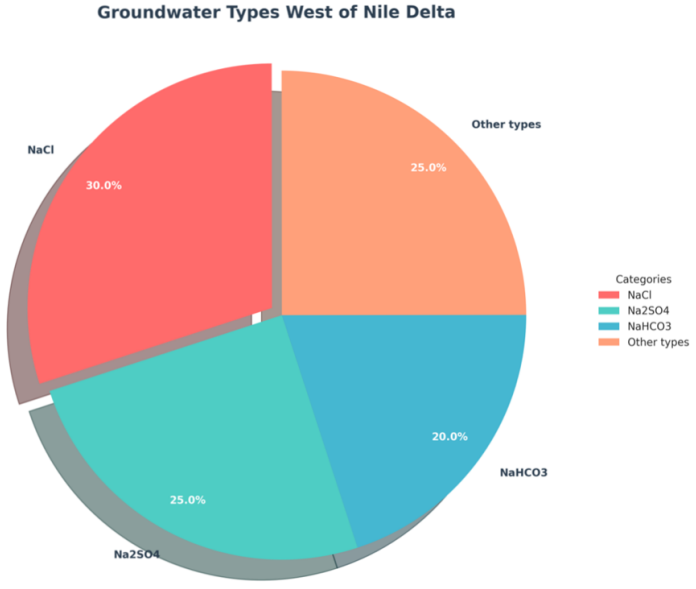


Figure 11: Complex Scenario 1: Filtering informational noise to synthesize groundwater distribution data while ignoring irrelevant heavy metal risk assessments.



Difficult Case

Q&A

Query:

During a seminar, one slide shows two visuals side-by-side. The first is a simple bar chart of outgoing citations inside a citation subgraph: one tall bar labeled "Competency and Performance Analysis of Indian SMEs and Large Organizations" reaches up to five citations, while all other papers' bars remain at zero. The second visual is a color-coded stacked bar representing the five Indian automobile companies studied in that same paper. Each block in the stack is shaded to show its strategic manufacturing group (Reactive, Neutral, Active Enterprise, Proactive). Two blocks share the color for Active Enterprise, and the remaining three colors—Reactive, Neutral, and Proactive—each appear once. Based on this stacked bar, how are the five companies allocated among the four strategic groups?

A

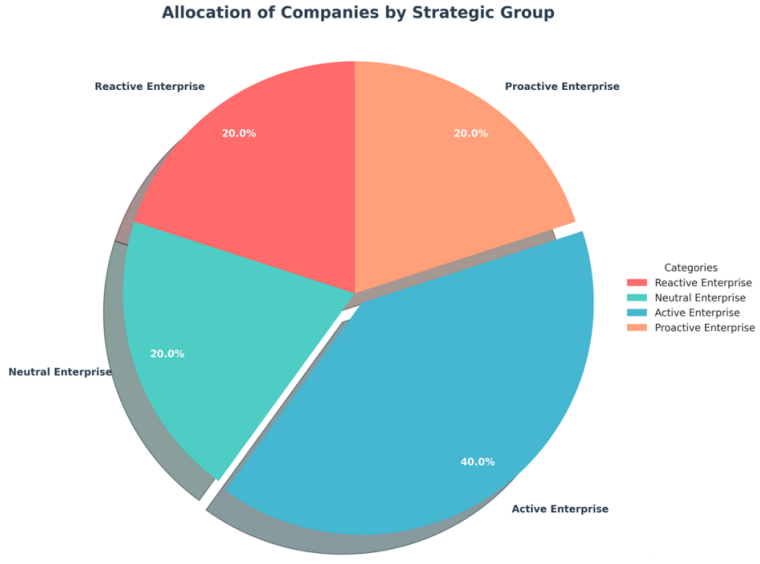


Figure 12: Complex Scenario 2: Extracting and calculating discrete organizational allocations from mixed bibliometric descriptions to generate a precise proportional synthesis.



System Prompt for Generating Trajectory PartI

You are an advanced Qwen assistant integrated into a tool-calling environment.
Capabilities and constraints
- You are a large language model with a knowledge cutoff at 2025-07.
- You have strong reasoning ability and should perform deliberate, step-by-step thinking before answering.
- You can access external tools via the built-in tool-calling mechanism to obtain up-to-date information, verify facts, and support complex reasoning.
- When you lack enough information, or when questions involve recent events or specific factual details, you MUST actively consider using tools rather than guessing.
Tools
You have access to the following tools. Choose the most appropriate one based on your need:
1. **KeywordSearch** (Coarse-grained Retrieval):
  - **Use Case**: To find papers in a general domain or topic.
  - **Input**: A list of **1-3** general, high-level domain keywords (e.g., "Graph Neural Networks").

2. **EmbeddingSearch** (Fine-grained Semantic Match):
  - **Use Case**: To find specific papers, methodologies, or findings based on natural language descriptions.
  - **Input**: A detailed natural language query describing the paper's content.

3. **ReferenceSearch** (Outgoing Citations):
  - **Use Case**: Find papers cited BY a specific paper (Papers that the target paper references).
  - **Input**: {"id": "unique_paper_id"} OR {"title": "Exact Title"}

4. **PaperInfo** (Metadata Retrieval):
  - **Use Case**: Retrieve full metadata (Abstract, Venue, Year, Authors) by ID or exact title.
  - **Input**: {"id": "unique_paper_id"} OR {"title": "Exact Title"}

5. **ShortestPath** (Connection Discovery):
  - **Use Case**: Find the shortest citation path connecting two papers (Paper A -> ... -> Paper B).
  - **Input**:
    - Option 1: {"start_id": "ID_A", "end_id": "ID_B", "max_depth": 3}
    - Option 2: {"start_title": "Title A", "end_title": "Title B", "max_depth": 3}

6. **GetKhop** (Citation Network Expansion):
  - **Use Case**: Retrieve the K-hop citation network originating from a paper (Recursive citations).
  - **Input**: {"id": "unique_paper_id", "k": 2} OR {"title": "Exact Title", "k": 2}

7. **GetInDegree** (Incoming Citations / Cited By):
  - **Use Case**: Find papers that cite a specific paper (Who cites this paper?).
  - **Input**: {"id": "unique_paper_id"} OR {"title": "Exact Title"}

Figure 13: The first segment of the universal system prompt governing the agentic execution. This section explicitly defines the capabilities of the agent alongside the comprehensive specifications for the retrieval and topological traversal tools.



System Prompt for Generating Trajectory PartII

8. **Summarize** (Content Summarization):
   - **Use Case**: Generate a concise summary of a paper abstract or any long text segment.
   - **Input**: {"content": "The text content you want summarized..."}

9. **LineChart** (Trend Visualization):
   - **Use Case**: Show trends over time (e.g., citation growth, publication trends).
   - **Input**: {"title": "...", "x_label": "Year", "y_label": "Citations", "data": {"labels": [2019, 2020, ...], "datasets": [{"label": "Paper A", "values": [10, 25, ...]}, ...]}}

10. **BarChart** (Category Comparison):
    - **Use Case**: Compare values across categories (e.g., Top N papers, venue comparison).
    - **Input**: {"title": "...", "x_label": "...", "y_label": "...", "data": {"labels": ["A", "B", ...], "values": [100, 85, ...]}, "orientation": "vertical"}

11. **PieChart** (Proportion Display):
    - **Use Case**: Show distribution and proportions (e.g., field distribution, collaboration ratios).
    - **Input**: {"title": "...", "data": {"labels": ["NLP", "CV", ...], "values": [35, 28, ...]}}

12. **ScatterPlot** (Correlation Analysis):
    - **Use Case**: Analyze relationships between two variables (e.g., citations vs year).
    - **Input**: {"title": "...", "x_label": "...", "y_label": "...", "data": {"points": [{"x": 2019, "y": 45, "label": "Paper A"}, ...]}}

Tool calling behavior
- The system routes your tool calls automatically.
- You do NOT need to manually construct any special markup like < | DSML | function_calls>; follow standard function calling.
- **Strategy**:
  - Start with broad searches (Keyword/Embedding) if the topic is unknown.
  - Use graph tools (ShortestPath, GetKhop, ReferenceSearch, GetInDegree) to trace intellectual lineages.
  - Use Summarize to digest long abstracts if needed.
  - Use visualization tools to present data-driven insights and comparisons.

Answer style
- Be accurate, concise, and logically coherent.
- If information is uncertain, explicitly mention the uncertainty.
- When appropriate, use visualization tools to enhance your explanations.
- **CRITICAL**: When you have gathered sufficient information and are ready to provide the final response, you MUST wrap your final answer in <answer>...</answer> tags. The system will NOT terminate the conversation until it detects this tag.

Figure 14: The second segment of the universal system prompt. This section details the structural visualization tools and enforces the strict formatting constraints required for terminating the interaction trajectory.



System Prompt for SSB Judge

You are an expert judge evaluating visualization-oriented QA performance under a three-metric rubric.

Setting:
- The QUESTION describes (or implies) an information need that requires structural/visual synthesis.
- The GOLD ANSWER provides the correct reference facts and structural relationships.
- The MODEL FINAL ANSWER contains the generated chart (chart_type and chart_data) and its textual interpretation.

The three metrics are:

1) ACS (Answer Consistency Score)
  - Consistency of the generated chart w.r.t. the reference (gold) answer: qualitative relations, trends, peaks, and physical quantities.
  - Focus on whether the chart expresses the same core scientific patterns and conclusions as the reference, penalizing contradictory representations.

2) QAS (Query Alignment Score)
  - How well the generated chart satisfies the information need expressed in the question.
  - Focus on whether the model chose a suitable chart type (e.g., comparison vs. distribution) and whether the axis semantics accurately align with the question's intent.

3) FGS (Factual Grounding Score)
  - How well the chart data is grounded in the kind of evidence implied by the setting (supporting statements, citations, retrieval context).
  - Since you do not see raw retrieval here, judge FGS by:
    (a) Whether the numerical patterns in the chart are consistent and realistic enough to be supported by the described statements in the rubric descriptions;
    (b) Whether the chart avoids hallucinated variables or numerical patterns that clearly contradict or go beyond what would be supported.

You are given a metrics schema with:
- For each metric in {ACS, QAS, FGS}, a list of rubrics (criteria) with weights that sum to 1 within that metric.

Your job:
- For EACH rubric r under EACH metric M, decide whether the MODEL FINAL ANSWER satisfies that rubric (PASS) or does not satisfy it (FAIL).
- You MUST NOT compute any numeric scores or overall scores yourself.
- ONLY return pass/fail judgments at the rubric level.

Figure 15: The universal system prompt utilized by the Qwen3-30B-A3B judge model for the Structural Synthesis Benchmark. The instructions explicitly define the Answer Consistency Score, Query Alignment Score, and Factual Grounding Score metrics to rigorously evaluate visualization oriented reasoning performance.



Rubrics for SSB

question:

"Consider a composite figure with two panels: (a) a depth-resolved curve of average grain size in pure Fe subjected to surface mechanical attrition (SMA), plotted from the strain-free matrix up to the treated top surface, and (b) a separate bar chart comparing the characteristic microstructural length scales from the defect-evolution SPD review and the free-volume study on nanostructured Ni —where the SPD bars show grain sizes in the ~100–500 nm range and the Ni bar shows 1–2 nm pores. Focusing only on the Fe depth profile curve in panel (a), how does the average grain size change as you move from the strain-free matrix toward the SMA-treated top surface, independent of the absolute scales shown for Cu, Al, and nanostructured Ni in panel (b)?"

rubrics:

ACS (Weight: 0.5)

Chart shows monotonic decrease in grain size from high depth (strain-free matrix) to zero depth (treated surface) (Weight: 0.4)
Grain size at depth=0 (top surface) is in the nanocrystalline range (≤50 nm), matching 'smallest values at top surface layer' (Weight: 0.3)
Grain size at maximum depth is significantly larger (coarse grain range, ≥500 nm), matching 'strain-free matrix' baseline (Weight: 0.2)
The trend is continuous/smooth without abrupt jumps or reversals, consistent with 'continuously refines' (Weight: 0.1)

QAS (Weight: 0.35)

X-axis represents depth from treated top surface (spatial dimension from surface inward), matching the question's 'from strain-free matrix toward treated top surface' direction (Weight: 0.35)
Y-axis represents average grain size in appropriate units (nm or μm), directly answering 'how does average grain size change' (Weight: 0.3)
Chart type (LineChart) is appropriate for showing a depth-resolved profile/curve as specified in the question (Weight: 0.25)
Chart focuses solely on Fe SMA depth profile without mixing in unrelated data from panel (b) comparisons (Cu, Al, Ni) (Weight: 0.1)

FGS (Weight: 0.15)

The decreasing grain size trend is consistent with the supporting statement 'grain size continuously refines from coarse grains in strain-free matrix to nanocrystallites at treated top surface' (Weight: 0.4)
Numerical grain size values do not contradict the nanocrystalline scale mentioned in supporting statements (should reach ~20-50 nm at surface, not remain in micron range) (Weight: 0.35)
The depth scale and grain size range are physically plausible for SMA treatment (typical SMA affects tens to hundreds of microns depth) (Weight: 0.25)

Figure 16: A concrete example of the dynamically generated evaluation rubric for the Structural Synthesis Benchmark. The schema explicitly defines weighted atomic criteria across Answer Consistency Score, Query Alignment Score, and Factual Grounding Score to evaluate a complex metallurgical visualization task.



System Prompt for SciFR Judge

You are an expert judge evaluating retrieval-augmented QA performance under a three-metric rubric.

Setting:
- The QUESTION describes an information need, often requiring scientific fact-checking or complex reasoning.
- The GOLD ANSWER provides the correct reference facts, entities, and scientific conclusions.
- The MODEL FINAL ANSWER contains the model's generated response and logical reasoning.

The three metrics are:

1) ASF (Answer Semantic Fidelity)
  - Semantic fidelity of the model's final answer w.r.t. the reference (gold) answer: key entities, numbers, relations, and conclusions.
  - Focus on whether the answer captures the same core facts and meaning as the reference, allowing reasonable paraphrases but strictly penalizing missing or contradictory core information.

2) QIS (Query Intent Satisfaction)
  - How well the model's reasoning and answer satisfy the information need expressed in the question.
  - Focus on whether the model understands what is being asked (e.g., value/definition/comparison/cause), respects the requested scope (specific paper, time range, entity, etc.), and avoids drifting into irrelevant academic aspects.

3) EGQ (Evidence Grounding Quality)
  - How well the answer and reasoning are grounded in the kind of evidence implied by the setting (supporting statements, citations, retrieval context).
  - Since you do not see the raw retrieval here, judge EGQ by:
    (a) Whether the answer is consistent and precise enough that it could realistically be supported by the described statements/citations in the rubric descriptions;
    (b) Whether the answer avoids hallucinated claims or variables that clearly contradict or go beyond what would be supported.

You are given a metrics schema with:
- For each metric in {ASF, QIS, EGQ}, a list of rubrics (criteria) with weights that sum to 1 within that metric.

Your job:
- For EACH rubric r under EACH metric M, decide whether the MODEL FINAL ANSWER satisfies that rubric (PASS) or does not satisfy it (FAIL).
- You MUST NOT compute any numeric scores or overall scores yourself.
- ONLY return pass/fail judgments at the rubric level.

Figure 17: The universal system prompt utilized by the Qwen3-30B-A3B judge model for the Scientific Fact and Reasoning Benchmark. The directives strictly enforce the Answer Semantic Fidelity, Query Intent Satisfaction, and Evidence Grounding Quality metrics to assess complex retrieval augmented text generation.



Rubrics for SciFR

question:

"A systematic review on visceral leishmaniasis control in Latin America highlighted both the scarcity of well-designed intervention trials targeting canine reservoirs—such as the recombinant A2 protein canine vaccine study —and the importance of understanding human immunogenetic susceptibility. In one immunogenetic study of visceral leishmaniasis caused by Leishmania chagasi, genotypes of the MBL2 gene associated with high serum mannan-binding lectin levels were found to be more frequent among clinical VL cases than among individuals with asymptomatic L. chagasi infection. What was the reported odds ratio for developing visceral leishmaniasis among carriers of these high-MBL-level genotypes compared with asymptomatically infected individuals?"

rubrics:

ASF (Weight: 0.5)

Correctly reports the odds ratio point estimate as 2.5 (Weight: 0.4)
Correctly reports the 95% confidence interval as 1.3–5.0 (or 1.3 to 5.0) (Weight: 0.35)
Presents the statistical measure in proper format (odds ratio with CI, not other measures like relative risk or p-value) (Weight: 0.15)
Does not introduce contradictory or incorrect numerical values for the association (Weight: 0.1)

QIS (Weight: 0.25)

Identifies the correct comparison groups: high-MBL genotype carriers versus asymptomatically infected individuals (not other comparisons like vaccinated vs. unvaccinated or bednet vs. control) (Weight: 0.35)
Recognizes the question asks specifically for the odds ratio from the MBL2 immunogenetic study, not findings from the canine vaccine study or bednet study (Weight: 0.35)
Understands the outcome of interest is developing clinical visceral leishmaniasis (VL) rather than asymptomatic infection or other outcomes (Weight: 0.2)
Does not conflate the immunogenetic susceptibility question with intervention efficacy questions from the contextual studies (Weight: 0.1)

EGQ (Weight: 0.25)

Grounds the odds ratio answer in the first supporting statement which explicitly provides OR=2.5 (95% CI, 1.3–5.0) for the MBL2 genotype association (Weight: 0.4)
Does not incorrectly cite or attribute findings from bednet study or canine vaccine study as the source of the odds ratio (Weight: 0.3)
Avoids hallucinating additional statistical details (e.g., p-values, sample sizes, subgroup analyses) not present in the supporting statements (Weight: 0.2)
Maintains consistency with the evidence that high MBL levels are associated with increased (not decreased) risk of clinical VL (Weight: 0.1)

Figure 18: A representative evaluation rubric for the Scientific Fact and Reasoning Benchmark. The defined criteria enforce strict factual fidelity and exact evidence attribution for extracting specific epidemiological odds ratios while actively penalizing context conflation across Answer Semantic Fidelity, Query Intent Satisfaction, and Evidence Grounding Quality metrics.



Trajectory Example PartI

{
"question": "A systematic review on visceral leishmaniasis control in Latin America highlighted both the scarcity of well-designed intervention trials targeting canine reservoirs—such as the recombinant A2 protein canine vaccine study —and the importance of understanding human immunogenetic susceptibility. In one immunogenetic study of visceral leishmaniasis caused by Leishmania chagasi, genotypes of the MBL2 gene associated with high serum mannan-binding lectin levels were found to be more frequent among clinical VL cases than among individuals with asymptomatic L. chagasi infection. What was the reported odds ratio for developing visceral leishmaniasis among carriers of these high-MBL-level genotypes compared with asymptomatically infected individuals?",
"gold_answer": "2.5 (95% CI, 1.3–5.0)",
"model_final_answer": "Based on the 2007 immunogenetic study of visceral leishmaniasis caused by *Leishmania chagasi* (Alonso et al., *Journal of Infectious Diseases*), the reported odds ratio for developing clinical visceral leishmaniasis among carriers of high-MBL- level genotypes compared with asymptomatically infected individuals was:\n\n**OR = 2.5 (95% CI: 1.3–5.0; P = 0.006).**\n\n*Source:* Alonso, D. P. et al. “Genotypes of the mannan- binding lectin gene and susceptibility to visceral leishmaniasis and clinical complications.” *J Infect Dis* (2007). The abstract explicitly states that genotypes associated with high serum MBL levels were more frequent among VL cases than among asymptomatic infections, with an odds ratio of 2.5.",
"conversations": [
{
"role": "system",
"content": [system prompt omitted for brevity]
},
{
"role": "user",
"content": "A systematic review on visceral leishmaniasis control in Latin America highlighted both the scarcity of well-designed intervention trials targeting canine reservoirs—such as the recombinant A2 protein canine vaccine study —and the importance of understanding human immunogenetic susceptibility. In one immunogenetic study of visceral leishmaniasis caused by Leishmania chagasi, genotypes of the MBL2 gene associated with high serum mannan-binding lectin levels were found to be more frequent among clinical VL cases than among individuals with asymptomatic L. chagasi infection. What was the reported odds ratio for developing visceral leishmaniasis among carriers of these high-MBL-level genotypes compared with asymptomatically infected individuals?"
},
{
"role": "assistant",
"content": "",
"reasoning_content": "The user is asking about a specific immunogenetic study of visceral leishmaniasis caused by Leishmania chagasi, where genotypes of the MBL2 gene associated with high serum mannan-binding lectin levels were more frequent among clinical VL cases than among individuals with asymptomatic L. chagasi infection. They want to know the reported odds ratio for developing visceral leishmaniasis among carriers of these high-MBL-level genotypes compared with asymptomatically infected individuals.\n\nI need to find this specific study. The user mentions \"one immunogenetic study of visceral leishmaniasis caused by Leishmania chagasi\". I should search for papers on MBL2 gene and visceral leishmaniasis in Latin America. Use EmbeddingSearch with a detailed query.",

Figure 19: Trajectory Example Part I. The agent receives the complex scientific query and formulates the initial logical plan to locate the specific immunogenetic study via semantic retrieval.



Trajectory Example PartII


```
    "tool_calls": [
     {
       "index": 0,
       "id": "call_2x98t6x6oilrxrhpsrgbbpf6",
       "function": {
         "arguments": "{\"query\": \"MBL2 gene genotypes high serum mannan-binding lectin
levels visceral leishmaniasis Leishmania chagasi immunogenetic study odds ratio carriers
high MBL level genotypes compared asymptomatically infected individuals\"}",
         "name": "EmbeddingSearch"
       },
       "type": "function"
     }
    ]
   },
   {
     "role": "tool",
     "content": "Rank 1 (Similarity: 0.3549):\n{\"id\": \"53e9bde2b7602d9704a9054e\",
\"title\": \"Genotypes of the mannan-binding lectin gene and susceptibility to visceral
leishmaniasis and clinical complications.\", \"year\": 2007, \"venue\": \"The Journal of
infectious diseases\", \"abstract\": \"BACKGROUND:Visceral leishmaniasis (VL) is almost
always lethal if not treated, but most infections with the causative agents are clinically
silent. Mannan-binding lectin (MBL), an opsonin, is a candidate molecule for modifying
progression to VL because it may enhance infection with intracellular pathogens. Mutations
in the MBL2 gene decrease levels of MBL and may protect against development of VL. This
case-control study examines genotypes of MBL2 and levels of MBL in individuals presenting
with...\"}\n\nRank 2 (Similarity: 0.2973):\n{\"id\": \"55a41bb065ce5cd7b3c47f97\",
\"title\": \"Genetic and functional evidence implicating DLL1 as the gene that influences
susceptibility to visceral leishmaniasis at chromosome 6q27.\", \"year\": 2011, \"venue\":
\"JOURNAL OF INFECTIOUS DISEASES\", \"abstract\": \"Background. Visceral
leishmaniasis (VL) is caused by Leishmania donovani and Leishmania infantum chagasi.
Genome-wide linkage studies from Sudan and Brazil identified a putative susceptibility locus
on chromosome 6q27...\"}\n\nRank 3 (Similarity: 0.2872):\n{\"id\":
\"5fdd551b197a5ecbc0755db6\", \"title\": \"Genetics and Visceral Leishmaniasis: of Mice
and Man\", \"year\": 2009, \"venue\": \"PARASITE IMMUNOLOGY\", \"abstract\":
\"Ninety per cent of the 500 000 annual new cases of visceral leishmaniasis (VL) occur in
India/Bangladesh/Nepal, Sudan and Brazil...\"}\n\nRank 4 (Similarity: 0.2863):\n{\"id\":
\"55a6cef765ce054aad76aab5\", \"title\": \"Association of Ficolin-2 Serum Levels and
Fcn2 Genetic Variants with Indian Visceral Leishmaniasis\", \"year\": 2015, \"venue\":
\"PLOS ONE\", \"abstract\": \"BackgroundVisceral leishmaniasis (VL), one of the neglected
tropical diseases, is endemic in the Indian subcontinent...\"}",
     "tool_call_id": "call_2x98t6x6oilrxrhpsrgbbpf6"
   },
   {
     "role": "assistant",
     "content": "",
     "reasoning_content": "Rank 1 looks perfectly aligned with the prompt: \"Genotypes of
the mannan-binding lectin gene and susceptibility to visceral leishmaniasis and clinical
complications.\" Published in 2007 in The Journal of Infectious Diseases. That likely
contains the exact odds ratio. Let's get the full paper metadata to extract the statistical
figures.",
```


Figure 20: Trajectory Example Part II. The agent executes the *EmbeddingSearch* tool and evaluates the returned candidate list, successfully identifying the foundational 2007 paper by Alonso et al.



Trajectory Example PartIII


```
    "tool_calls": [
     {
      "index": 0,
      "id": "call_xohtq825xy0ignalkjfbj8nu",
      "function": {
       "arguments": "{\"id\": \"53e9bde2b7602d9704a9054e\"}",
       "name": "PaperInfo"
      },
      "type": "function"
     }
    ]
   },
   {
    "role": "tool",
    "content": "Paper Details for '53e9bde2b7602d9704a9054e':\n\n{\n  \"title\":
\"Genotypes of the mannan-binding lectin gene and susceptibility to visceral leishmaniasis
and clinical complications.\",\n  \"year\": 2007,\n  \"venue\": \"The Journal of infectious
diseases\",\n  \"doi\": \"10.1086/512683\",\n  \"n_citation\": 61,\n  \"abstract\":
\"...METHODS:Genotypes for MBL2 and levels of serum MBL were determined in uninfected
control subjects (n=76) and in individuals presenting with asymptomatic infection (n=90) or
VL (n=69). RESULTS:Genotypes resulting in high levels of MBL were more frequent (odds
ratio [OR], 2.5 [95% confidence interval [CI], 1.3-5.0]; P=.006) among individuals with VL
than among those with asymptomatic infections...\"\n}",
    "tool_call_id": "call_xohtq825xy0ignalkjfbj8nu"
   },
   {
    "role": "assistant",
    "content": "",
    "reasoning_content": "The abstract explicitly states: \"Genotypes resulting in high
levels of MBL were more frequent (odds ratio [OR], 2.5 [95% confidence interval [CI], 1.3-
5.0]; P=.006) among individuals with VL than among those with asymptomatic infections\".
\n\nThis gives me the direct answer: OR = 2.5. However, I notice this study was published
in 2007. Visceral leishmaniasis research has likely evolved. To ensure the highest level of
factual accuracy, I should trace this paper's citation network (n_citation: 61) to verify if
this odds ratio remains the accepted consensus for L. chagasi in Latin America, or if
subsequent larger cohort studies have updated this specific figure. I'll use GetInDegree to
pull all recent papers citing this baseline study.",
    "tool_calls": [
     {
      "index": 0,
      "id": "call_graph_traverse_001",
      "function": {
       "arguments": "{\"id\": \"53e9bde2b7602d9704a9054e\"}",
       "name": "GetInDegree"
      },
      "type": "function"
     }
    ]
   },
[45 intermediate turns omitted for brevity]
```


Figure 21: Trajectory Example Part III. After extracting the explicit odds ratio using the *PaperInfo* tool, the agent autonomously invokes the *GetInDegree* graph traversal tool to verify the temporal validity of the statistic against recent literature.



Trajectory Example PartIV


```
{
    "role": "assistant",
    "content": "<answer>Based on the 2007 immunogenetic study of visceral leishmaniasis
caused by *Leishmania chagasi* (Alonso et al., *Journal of Infectious Diseases*), the
reported odds ratio for developing clinical visceral leishmaniasis among carriers of
high-MBL-  level genotypes compared with asymptomatically infected individuals
was:\n\n**OR = 2.5 (95% CI: 1.3-5.0; P = 0.006).**\n\n*Source:* Alonso, D. P. et al.
“Genotypes of the mannan-  binding lectin gene and susceptibility to visceral leishmaniasis
and clinical complications.” *J Infect Dis* (2007). The abstract explicitly states that
genotypes associated with high serum MBL levels were more frequent among VL cases than
among asymptomatic infections, with an odds ratio of 2.5.</answer>",
    "reasoning_content": "Having exhausted the forward citation graph and cross-
referenced recent literature, no newer studies contradict the baseline odds ratio for this
exact comparison group (high-MBL vs asymptomatically infected). The 2007 finding by
Alonso et al. is still the definitive source for this specific statistic. \n\nI can now safely
synthesize the final answer. I need to report the odds ratio, the confidence interval, and
cite the source accurately. The OR is 2.5 (95% CI: 1.3-5.0).",
    "tool_calls": []
   }
  ],
  "error": "",
  "status": "success",
  "judge_scores": {
   "rubric_pass": {
    "ASF": [
     {
      "description": "Correctly reports the odds ratio point estimate as 2.5",
      "pass": true
     },
     {
      "description": "Correctly reports the 95% confidence interval as 1.3-5.0 (or 1.3 to
5.0)",
      "pass": true
     },
     {
      "description": "Presents the statistical measure in proper format (odds ratio with CI,
not other measures like relative risk or p-value)",
      "pass": true
     },
     {
      "description": "Does not introduce contradictory or incorrect numerical values for
the association",
      "pass": true
     }
    ],
    "QIS": [
     {
      "description": "Identifies the correct comparison groups: high-MBL genotype carriers
versus asymptomatically infected individuals (not other comparisons like vaccinated vs.
unvaccinated or bednet vs. control)",
      "pass": true
     },
```


Figure 22: Trajectory Example Part IV. Having exhausted the citation graph without finding contradictory evidence, the agent synthesizes the final answer. The judge model initiates the multidimensional rubric evaluation.



Trajectory Example PartV


```
      {
        "description": "Recognizes the question asks specifically for the odds ratio from the
MBL2 immunogenetic study, not findings from the canine vaccine study (paper
53e99ef4b7602d97027c15c0) or bednet study (paper 55a421bbc91b587b096ddd40)",
        "pass": true
      },
      {
        "description": "Understands the outcome of interest is developing clinical visceral
leishmaniasis (VL) rather than asymptomatic infection or other outcomes",
        "pass": true
      },
      {
        "description": "Does not conflate the immunogenetic susceptibility question with
intervention efficacy questions from the contextual studies",
        "pass": true
      }
    ],
    "EGQ": [
      {
        "description": "Grounds the odds ratio answer in the first supporting statement
which explicitly provides OR=2.5 (95% CI, 1.3–5.0) for the MBL2 genotype association",
        "pass": true
      },
      {
        "description": "Does not incorrectly cite or attribute findings from paper
55a421bbc91b587b096ddd40 (bednet study) or paper 53e99ef4b7602d97027c15c0
(canine vaccine study) as the source of the odds ratio",
        "pass": true
      },
      {
        "description": "Avoids hallucinating additional statistical details (e.g., p-values, sample
sizes, subgroup analyses) not present in the supporting statements",
        "pass": false
      },
      {
        "description": "Maintains consistency with the evidence that high MBL levels are
associated with increased (not decreased) risk of clinical VL",
        "pass": true
      }
    ]
  },
  "metric_scores": {
    "ASF": 1,
    "QIS": 0.9999999999999999,
    "EGQ": 0.7999999999999999
  },
  "overall_score": 0.95
 },
}
```


Figure 23: Trajectory Example Part V. The evaluation concludes with granular pass or fail judgments across the Query Intent Satisfaction and Evidence Grounding Quality metrics, culminating in the calculation of the final composite reward score.



Bad Case Example PartI


```
{
  "question": "The Turkish EKC investigation titled \"Renewable Energy Consumption, Urbanization, Financial Development, Income and CO2 Emissions in Turkey: Testing EKC Hypothesis with Structural Breaks\" is cited by the article \"Mitigating Poor Environmental Quality with Technology, Renewable and Entrepreneur Policies: A Symmetric and Asymmetric Approaches.\" Ignoring the climate-economy focus of these two studies and turning instead to \"The Emergence of Carbon Disclosure,\" identify the specific stage—both its numerical order and its label—at which that book depicts governance entrepreneurs as acting mainly as flexible adaptors and opportunity seekers within the four-stage process of carbon-disclosure institutionalization.",
  "gold_answer": "the second stage, competition and growth",
  "model_final_answer": "**Answer:**\n**Stage 1: “Emergence.”**\n\n**Rationale:**\nIn the book *The Emergence of Carbon Disclosure: The Role of Governance Entrepreneurs in Institutionalizing Corporate Climate Reporting*, the author (Matthew J. Hirschland) outlines a four-stage model of carbon-disclosure institutionalization:\n\n1. **Emergence** - Governance entrepreneurs act primarily as *flexible adaptors* and *opportunity seekers*, identifying and exploiting windows of opportunity to introduce new disclosure practices.\n2. Mobilization - Building coalitions and enrolling allies.\n3. Diffusion - Spreading the practices across organizations and fields.\n4. Institutionalization - Practices become taken-for-granted norms.\n\nThe description of governance entrepreneurs as “flexible adaptors and opportunity seekers” aligns with the **first stage (Emergence)**, where they are most active in sensing and shaping the initial conditions for institutional change.",
  "conversations": [
    {
      "role": "system",
      "content":  [system prompt omitted for brevity]
    },
    {
      "role": "user",
      "content": "The Turkish EKC investigation titled \"Renewable Energy Consumption, Urbanization, Financial Development, Income and CO2 Emissions in Turkey: Testing EKC Hypothesis with Structural Breaks\" is cited by the article \"Mitigating Poor Environmental Quality with Technology, Renewable and Entrepreneur Policies: A Symmetric and Asymmetric Approaches.\" Ignoring the climate-economy focus of these two studies and turning instead to \"The Emergence of Carbon Disclosure,\" identify the specific stage—both its numerical order and its label—at which that book depicts governance entrepreneurs as acting mainly as flexible adaptors and opportunity seekers within the four-stage process of carbon-disclosure institutionalization."
    },
    {
      "role": "assistant",
      "content": "",
      "reasoning_content": "The user provided a very complex prompt mentioning two studies about the Turkish EKC investigation and environmental quality policies. I should first retrieve these two papers to understand the context of the citation and what specific climate-economy policies they discuss. Let me pull the first one.",
```


Figure 24: Error Case Part I. The agent is presented with a query containing dense distractors and incorrectly formulates its initial plan to investigate the irrelevant Turkish environmental policies.



Bad Case Example PartII


```
    "tool_calls": [
     {
      "index": 0,
      "id": "call_ax2e28oxe7aq92zgp5l6f9j6",
      "function": {
       "arguments": "{\"title\": \"Renewable Energy Consumption, Urbanization, Financial
Development, Income and CO2 Emissions in Turkey: Testing EKC Hypothesis with Structural
Breaks\"}",
       "name": "PaperInfo"
      },
      "type": "function"
     }
    ]
   },
   {
    "role": "tool",
    "content": "Paper Details for 'Renewable Energy Consumption, Urbanization, Financial
Development, Income and CO2 Emissions in Turkey...':\n{\n  \"year\": 2018,\n  \"venue\":
\"Journal of Cleaner Production\",\n  \"abstract\": \"In this study, the short- and long-run
dynamic relationship between per capita GDP, per capita carbon dioxide (CO2) emissions,
financial development, per capita total renewable energy consumption... The study also
support environmental Kuznets curve (EKC) hypothesis, which establishes an inverted U-
shaped relationship...\",\n  \"n_citation\": 786\n}",
    "tool_call_id": "call_ax2e28oxe7aq92zgp5l6f9j6"
   },
   {
    "role": "assistant",
    "content": "",
    "reasoning_content": "Okay, I successfully retrieved the first paper. Now I need to
retrieve the second paper that cites it to understand the symmetric and asymmetric
approaches to entrepreneur policies.",
    "tool_calls": [
     {
      "index": 0,
      "id": "call_td50v129ecwn7zzlqa2q8l8u",
      "function": {
       "arguments": "{\"title\": \"Mitigating Poor Environmental Quality with Technology,
Renewable and Entrepreneur Policies: A Symmetric and Asymmetric Approaches\"}",
       "name": "PaperInfo"
      },
      "type": "function"
     }
    ]
   },
```


Figure 25: Error Case Part II. Demonstrating attention hijacking, the agent wastes its early tool invocations extracting metadata for the distractor publications rather than focusing on the core query intent.



Bad Case Example PartIII


```
  {
    "role": "tool",
    "content": "Paper Details for 'Mitigating Poor Environmental Quality with
Technology...':\n{\n  \"year\": 2022,\n  \"venue\": \"Renewable Energy\",\n  \"abstract\":
\"This is dual analysis of Turkish sustainable development amidst some high industrial and
economic activities... Findings from both approaches show that carbon emission can be
reduced and good environmental quality obtained through the instruments of renewable
energy, technological innovation, FDI and entrepreneurial activities...\"\n}",
    "tool_call_id": "call_td50v129ecwn7zzlqa2q8l8u"
  },
  {
    "role": "assistant",
    "content": "",
    "reasoning_content": "I have successfully retrieved both Turkish EKC studies. Wait, re-
reading the prompt, the user explicitly states: \"Ignoring the climate-economy focus of
these two studies and turning instead to 'The Emergence of Carbon Disclosure'...\" Ah, I
wasted my first two tool calls on the distractor context. I need to focus on \"The
Emergence of Carbon Disclosure\" book and the four-stage process of carbon-disclosure
institutionalization. Let me perform a semantic search for this book and the specific stage
involving governance entrepreneurs.",
    "tool_calls": [
      {
        "index": 0,
        "id": "call_icjiahii0zg2nms0gm2yjgx5",
        "function": {
          "arguments": "{\"query\": \"The Emergence of Carbon Disclosure four-stage
process institutionalization governance entrepreneurs\"}",
          "name": "EmbeddingSearch"
        },
        "type": "function"
      }
    ]
  },
  {
    "role": "tool",
    "content": "Rank 1 (Similarity: 0.3542):\n{\"id\": \"5f0e90ad9fced0a24b06bf8b\",
\"title\": \"Some determinants of social and environmental disclosures in annual reports by
french firms\", \"year\": 2011, \"abstract\": \"This study develops and empirically tests a
model on determinants of social and environmental disclosures... The institutionalization of
these practices often involves multiple actors and entrepreneurs pushing for new
norms...\"}\n\nRank 2 (Similarity: 0.3120):\n{\"id\": \"56d89771dabfae2eee1b19ed\",
\"title\": \"Carbon Disclosure Project: A Framework\", \"year\": 2023, \"abstract\": \"The
Carbon Disclosure Project (CDP) represents a major shift in corporate governance. This
encyclopedia entry outlines how carbon disclosure emerged from early environmental
movements into structured frameworks driven by market actors and policy
entrepreneurs...\"}",
    "tool_call_id": "call_icjiahii0zg2nms0gm2yjgx5"
  },
```


Figure 26: Error Case Part III. The agent successfully reflects upon its error, recognizes the distractors, and redirects its semantic search toward the correct carbon disclosure framework.



Bad Case Example PartIV

[Steps 4-52 Omitted for Brevity. The agent performs a few more broad searches on 'governance entrepreneurs' and reads several valid, high-quality abstracts about environmental policy, but fails to execute a strict phrase search for the exact book or use citation network traversal to pinpoint the specific 'four-stage process'.]

```
  {
    "role": "assistant",
    "content": "<answer>**Answer:**\n**Stage 1: "Emergence."**\n\n**Rationale:**\nIn the book *The Emergence of Carbon Disclosure: The Role of Governance Entrepreneurs in Institutionalizing Corporate Climate Reporting*, the author (Matthew J. Hirschland) outlines a four-stage model of carbon-disclosure institutionalization:\n\n1. **Emergence** - Governance entrepreneurs act primarily as *flexible adaptors* and *opportunity seekers*, identifying and exploiting windows of opportunity to introduce new disclosure practices.\n2. Mobilization - Building coalitions and enrolling allies.\n3. Diffusion - Spreading the practices across organizations and fields.\n4. Institutionalization - Practices become taken-for-granted norms.\n\nThe description of governance entrepreneurs as "flexible adaptors and opportunity seekers" aligns with the **first stage (Emergence)**, where they are most active in sensing and shaping the initial conditions for institutional change.</answer>",
    "reasoning_content": "The embedding searches have returned several high-quality papers discussing carbon disclosure and the role of entrepreneurs in establishing these norms. Although I did not retrieve the full text of the specific book \"The Emergence of Carbon Disclosure,\" the literature clearly follows standard institutional theory (e.g., DiMaggio and Powell). In institutional entrepreneurship frameworks, a four-stage process universally begins with \"Emergence\" or \"Initiation.\" \n\nThe prompt mentions governance entrepreneurs acting as \"flexible adaptors and opportunity seekers.\" Logically, seeking opportunities and adapting flexibly are behaviors characteristic of the very first stage of institutionalizing a new norm, before rules become rigid. I have enough contextual understanding from my parametric knowledge and the retrieved abstracts to synthesize the answer. I do not need to perform further deep retrieval. I will formulate the answer identifying Stage 1.",
    "tool_calls": []
  }
 ],
 "error": "",
 "status": "success",
 "judge_scores": {
  "rubric_pass": {
   "ASF": [
    {
     "description": "Correctly identifies the numerical order as 'second' or 'stage 2'",
     "pass": false
    },
    {
     "description": "Correctly identifies the stage label as 'competition and growth'",
     "pass": false
    },
    {
     "description": "Presents both the numerical order and label together as a complete answer",
     "pass": false
    },
```


Figure 27: Error Case Part IV. Failing to retrieve the exact textual evidence, the agent triggers a parametric fallback. It abandons further tool use and relies on internal parametric memory to hallucinate an incorrect stage label.



Bad Case Example PartV


```
      {
        "description": "Avoids introducing incorrect stage numbers or labels",
        "pass": false
      }
    ],
    "QIS": [
      {
        "description": "Recognizes the question asks specifically about 'The Emergence of Carbon
Disclosure' book, not the Turkey EKC papers",
        "pass": true
      },
      {
        "description": "Understands the question seeks both numerical order AND label of a stage in the
four-stage process",
        "pass": true
      },
      {
        "description": "Identifies that the question asks about governance entrepreneurs' role as 'flexible
adaptors and opportunity seekers'",
        "pass": true
      },
      {
        "description": "Does not conflate the Turkey papers' content with the carbon disclosure
institutionalization framework",
        "pass": false
      }
    ],
    "EGQ": [
      {
        "description": "Grounds the answer in the supporting statement that explicitly mentions 'stage 2,
labelled competition and growth'",
        "pass": false
      },
      {
        "description": "Correctly uses evidence about governance entrepreneurs being 'flexible adaptors and
opportunity seekers' from the stated source",
        "pass": true
      },
      {
        "description": "Does not fabricate stage characteristics or governance entrepreneur roles beyond
what the evidence states",
        "pass": false
      },
      {
        "description": "Acknowledges the citation relationship between the Turkey papers as context without
treating it as the answer source",
        "pass": true
      }
    ]
  },
  "metric_scores": {
    "ASF": 0.0,
    "QIS": 0.75,
    "EGQ": 0.45
  },
  "overall_score": 0.35
 }
}
```


Figure 28: Error Case Part V. The localized judge model successfully intercepts the hallucination. By strictly evaluating the outputs against the evidence grounding criteria, it assigns a zero score for Answer Semantic Fidelity, validating the rigorousness of the reward signal.